

Building Applications with OpsDB Application Architecture

A Construction Reference for the OpsDB Application Platform

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Abstract

OpsDB is a governed data substrate that unifies schema-driven validation, five-layer authorization, versioned state, change management with approval routing, append-only audit logging, structured search, and bounded automation into a single ten-step gate pipeline. Built in Go on Postgres and released under the MIT license, the compiled infrastructure contains zero application-domain types — all domain knowledge lives in YAML schema files and data rows interpreted by fixed mechanisms. An application built on OpsDB is called an AppDB. The developer writes schema YAML declaring entities, fields, types, constraints, and relationships. Running the schema loader produces a complete backend with validated writes, access control, version history, change management, audit trail, and search — without writing endpoint handlers, validation code, authorization logic, or audit infrastructure. Backend logic is expressed as runners — small programs following a three-phase read-act-write pattern through a shared library suite that handles all infrastructure concerns. A closed constraint vocabulary of nine types, three modifiers, six constraints, sixteen operations, and ten gate steps means security derives from structural limitation and the marginal cost of new application behavior approaches zero.

This paper enumerates 47 construction methods for building applications on OpsDB, covering domain analysis, schema design, policy configuration, runner implementation, frontend integration, and AI-assisted development. The methods apply across a spectrum from personal data platforms to enterprise backends under regulatory oversight, composing according to architecture position: primary backend for governed-state-dominant ap-

plications, split backend alongside specialized hot-path systems, or operational wrapper managing configuration and compliance around processing-dominant systems.

1. Development Lifecycle

Building an application on OpsDB follows a sequence that differs from conventional development at every stage. The sequence is: domain analysis, schema design, loader run, API verification, runner design, runner implementation, frontend integration. Each stage produces artifacts that the next stage consumes. No stage involves writing validation code, authorization logic, versioning systems, audit infrastructure, or API endpoint handlers.

1.1 The Artifact Set A complete AppDB application consists of four artifact types.

Schema YAML files. These declare every entity type, field, constraint, relationship, and governance configuration in the application. The schema is the source of truth for what data exists and what rules govern it. A typical governed-state-dominant application has 20–80 entity YAML files totaling 2–4K lines.

Runner specifications. These are OpsDB entities with typed JSON payloads that declare what each runner does, what it can access, what bounds it operates within, and what report keys it can write. Runner specs are governed data — versioned, change-managed, and auditable.

Runner code. Small programs — typically 150–300 lines each — that follow the get-act-set pattern using the shared library suite. A typical application has 3–8 runners. Each handles one domain-specific concern: pulling external data, reconciling state, computing derived quantities, dispatching notifications, generating reports, or producing AI-derived observations.

Frontend code. A thin consumer of the OpsDB API. It authenticates users via SSO, translates user actions into API calls, and renders responses. It has no database, no validation logic, no authorization logic. It renders what the API returns, including field-level omissions when access classification hides sensitive data. The frontend is conventional web or mobile development — React, Swift, Kotlin, server-rendered templates — and is not covered in depth here because OpsDB does not change how frontends are built, only what they connect to.

1.2 The Development Sequence **Step 1: Domain analysis.** Decompose the domain into entities, relationships, lifecycles, policies, schedules, external integrations, and hot-path processing (if any). The output is a domain map — a structured inventory of everything the application manages. This step is intellectual work, not code. It uses the comprehensive slicing method from the ops book: subdivide the domain preserving the whole, no information loss.

Step 2: Architecture position. Determine where OpsDB sits. Most business applications are governed-state-dominant — OpsDB is the entire backend. Applications with specialized processing needs identify the hot path and plan the runner bridge pattern. The taxonomy’s application suitability matrix provides the reference.

Step 3: Schema design. Translate the domain map into YAML files. Each entity type gets a file. Fields, types, constraints, relationships, governance fields, and discriminator patterns are declared. The schema conventions — singular names, lowercase underscores, hierarchical prefixes, FK naming, datetime suffixes, boolean prefixes — apply uniformly. This step typically takes hours to days, not weeks.

Step 4: Loader run. Run the schema loader against the YAML files. The loader parses, validates structural correctness, checks naming conventions, verifies foreign key targets exist, generates database DDL, creates tables with all reserved fields, and populates schema metadata tables. If the loader rejects the schema, the error identifies the specific violation. Fix the YAML file and re-run. When the loader succeeds, the API serves every declared entity with full properties — validation, authorization, versioning, change management, audit, search, retention.

Step 5: API verification. Write seed data through the API. Create entities, search them, update them through change sets, verify version history, check that constraints reject invalid data, confirm that access classification hides the right fields. This is testing — but the testing is against data declarations, not against code you wrote. The gate pipeline is the test runner. Every constraint violation you trigger confirms the schema is correct.

Step 6: Runner design. For each piece of backend logic the application needs — external integration, derived computation, notification dispatch, reconciliation, evidence production, AI observation — design a runner. Identify inputs, outputs, gating mode, trigger, bounds, and idempotency semantics. Write the runner spec as an OpsDB entity.

Step 7: Runner implementation. Write each runner using the library suite. The get phase reads from OpsDB and external sources. The act phase performs the domain-specific computation or integration. The set phase writes results back through the API. The library suite handles authentication, retry, circuit breaking, logging, correlation ID propagation, and scope enforcement. The runner handles only the domain decision.

Step 8: Frontend integration. Connect the frontend to the OpsDB API. Implement SSO authentication. Map user actions to API operations — search for list views, get_entity for detail views, submit_change_set for mutations, get_entity_history for activity feeds. Render approval status for pending change sets. Handle draft mode for fluid editing use cases.

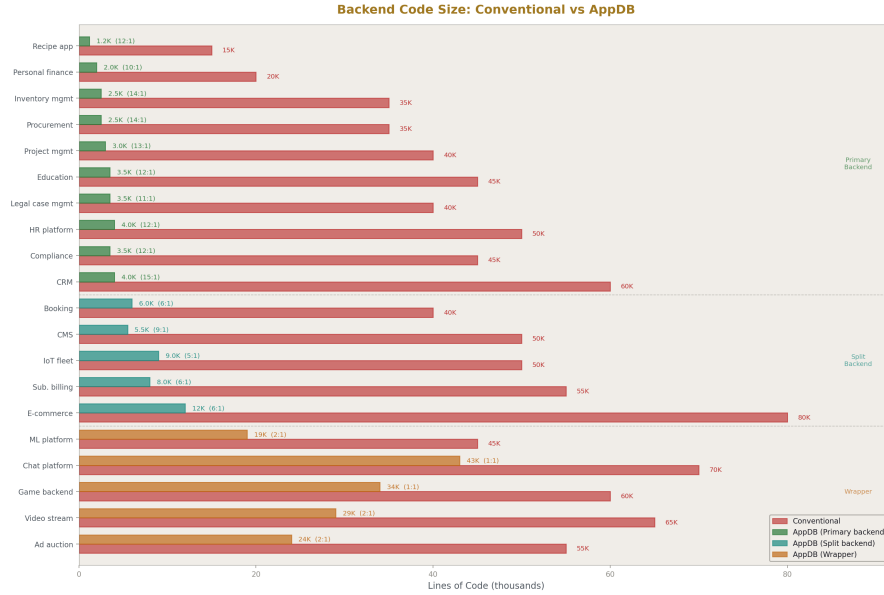


Figure 1: Fig. 1: Backend Code Size — conventional vs AppDB across 20 application types, showing 8–15x reduction for governed-state apps.

2. Domain Analysis Method

Domain analysis produces the input for schema design. The method is systematic: enumerate exhaustively, classify each element, and identify relationships before writing any YAML. This is comprehensive design — build from the whole, not from accumulated parts.

2.1 Entity Enumeration List every noun in the domain that has identity, attributes, and lifecycle. A project management domain: project, task, milestone, comment, label, attachment reference, team, member, role. A billing domain: customer, account, plan, subscription, invoice, line item, payment, credit, usage record. A healthcare domain: patient, provider, encounter, diagnosis, medication, order, result, care plan, insurance.

For each entity, ask: does this thing have its own identity (it is something, not just a property of something else)? Does it have attributes beyond the reference to its parent? Can it exist independently or does it exist only in context of another entity? Things with independent identity are entities. Things that are properties of other entities are fields on those entities. Things that represent relationships between entities are bridge tables.

2.2 Field Enumeration For each entity, list every attribute. For each attribute, determine type (from the nine available: int, float, varchar, text, boolean, datetime, date,

json, enum), constraints (numeric ranges, string lengths, enum value sets, foreign key references), and modifiers (nullable, default, unique).

The closed vocabulary constrains this step productively. You cannot declare a regex constraint, so you do not need to design one. You cannot declare a conditional constraint, so cross-field invariants go to policy rows. You cannot declare computed defaults, so derived values go to runners. The vocabulary forces clean separation between what the schema handles (structural validation) and what runners handle (domain computation).

2.3 Relationship Classification Relationships between entities fall into five patterns.

Direct ownership. A task belongs to a project. The task has a `project_id` foreign key. The relationship is mandatory (not nullable) and hierarchical — deleting the project affects its tasks.

Optional reference. A task may reference an assignee. The task has a `assignee_user_id` foreign key that is nullable. The relationship is informational, not structural.

Many-to-many bridge. A task can have multiple labels. A label can apply to multiple tasks. A `task_label` bridge table has `task_id` and `label_id` foreign keys, both mandatory. The bridge table may have its own fields — `applied_time`, `applied_by_user_id`.

Polymorphic bridge. A comment can be attached to a task, a project, or a milestone. One bridge table per source-target pair: `task_comment`, `project_comment`, `milestone_comment`. Each bridge has the comment FK and the target FK. This preserves foreign key integrity — no polymorphic FK columns that could reference multiple tables.

Self-referential hierarchy. A task can have subtasks. The task has a nullable `parent_task_id` foreign key referencing its own table. Hierarchy traversal uses `get_dependencies` with the parent chain join path.

2.4 Lifecycle Identification Some entities progress through states with rules about valid transitions. Identify which entities have lifecycles and enumerate the states and transitions.

A task: `draft` → `open` → `in_progress` → `review` → `done` → `archived`. Valid transitions: `draft`→`open`, `open`→`in_progress`, `in_progress`→`review`, `review`→`done`, `review`→`in_progress` (returned for rework), `done`→`archived`, any→`draft` (reset). Invalid transitions: `done`→`in_progress` (must go through `draft`), `archived`→anything (terminal).

A subscription: `trial` → `active` → `past_due` → `suspended` → `canceled` → `expired`. Each transition has rules: `trial`→`active` requires payment method, `active`→`past_due` triggered by failed payment, `past_due`→`suspended` after configurable grace period, `canceled` is human-initiated, `expired` is system-triggered after cancellation period.

The states become an enum field on the entity. The valid transitions become policy rows evaluated at gate step 5. The transition rules that involve computation — “grace period

elapsed,” “payment method valid” — become runner logic that proposes change sets when conditions are met.

2.5 Policy Enumeration For each entity and relationship, identify the governance requirements.

Approval rules. Which changes require human approval? Typically: changes to financial data, changes to access control configuration, changes to compliance-relevant fields, changes proposed by automated systems operating above a risk threshold. Low-risk changes — task status updates, comment additions, label assignments — auto-approve.

Access control. Which entities are visible to which roles? Which fields within an entity are sensitive? Patient diagnosis is visible only to the care team. Salary data is visible only to HR and the employee. Financial totals are visible only to billing administrators. These map to `_requires_group` (entity-level) and `_access_classification` (field-level) in the schema.

Retention. How long does version history persist for each entity type? Compliance-relevant entities may require 7-year retention. Operational telemetry may require 30-day retention. Personal notes may have no retention policy — keep forever on personal hardware.

Change management. Emergency review windows, segregation of duties requirements, mandatory review periods. These are policy rows, not schema declarations.

2.6 Schedule Enumeration Identify everything that recurs, expires, or has deadlines.

Recurring operations. Credential rotation every 90 days. Compliance audit quarterly. Invoice generation monthly. Report generation weekly. Backup verification daily. Each becomes a schedule entity with a typed payload — cron expression, rate-based, or calendar-anchored — connected to a runner spec.

Deadlines. Task due dates. Contract renewal dates. Certificate expiration dates. Each is a datetime field on the entity. A verifier runner reads approaching deadlines and produces evidence records or triggers notifications.

Event-triggered schedules. “Run compliance scan 24 hours after any policy change.” The schedule entity uses the event-triggered payload type, referencing the entity type and field change that triggers it.

2.7 External Integration Enumeration Identify every external system the application interacts with.

Authorities. External systems that are the source of truth for some data. A payment processor is the authority for payment status. An identity provider is the authority for user identity. A cloud provider is the authority for resource state. Each authority gets an authority entity in OpsDB with a typed JSON payload describing the connection parameters.

Puller targets. For each authority, what data does the application need from it? Payment status, user groups, resource inventory. Each becomes a puller runner that reads from the authority and writes to observation cache tables.

Push targets. External systems that consume configuration from the application. An API gateway that reads route definitions. A CDN that reads content configuration. A notification service that receives dispatch requests. Each becomes a configuration push runner.

2.8 Hot Path Identification Determine whether the application has processing that cannot go through the gate pipeline.

The gate pipeline adds latency — authentication, five-layer authorization, schema validation, bound validation, policy evaluation, versioning, change management routing, audit logging, execution, response construction. For governed-state operations at human pace, this latency is invisible. For operations requiring sub-millisecond response, high-frequency writes, or compute-intensive processing, the pipeline is too heavy.

If the application has such processing, identify it, bound it, and plan the runner bridge. The hot path gets its own system. OpsDB governs everything around it. Runners push configuration to the hot path system and pull results back. The hot path system operates independently of OpsDB availability.

Most applications have no hot path. The gate pipeline handles human-pace CRUD, search, and change management without perceptible latency. The developer should not assume a hot path exists — verify through measurement, not assumption.

3. Schema Construction Methods

Schema construction translates the domain analysis into YAML files that the loader consumes. Each method addresses a specific schema design pattern.

3.1 Entity Design An entity YAML file declares the entity's name, fields, relationships, governance configuration, and notes. The minimal structure:

```
name: task

domain: project_management

versioned: true

notes: |
```

```
  A unit of work within a project. Lifecycle managed through
```

status enum with transitions enforced by policy rows.

```
fields:
  - name: title
type: varchar
length: 255
nullable: false
  - name: description
type: text
nullable: true
  - name: status
type: enum
values: [draft, open, in_progress, review, done, archived]
nullable: false
default: draft
  - name: priority
type: int
min: 1
max: 5
nullable: false
default: 3
  - name: due_date
type: date
nullable: true
```

```

    - name: estimated_hours
type: float
min: 0.0
max: 10000.0
nullable: true

    - name: project_id
type: foreign_key
references: project
nullable: false

    - name: assignee_user_id
type: foreign_key
references: ops_user
nullable: true

    - name: _requires_group
type: foreign_key
references: ops_group
nullable: true

    - name: _access_classification
type: foreign_key
references: data_classification
nullable: true

```

Every field has an explicit type from the nine available. Every numeric field has explicit bounds. Every varchar has an explicit length. Every enum has an explicit value set. Every foreign key references a specific target table. The loader rejects anything outside this vocabulary.

The reserved fields — `id`, `created_time`, `updated_time`, `is_active`, `created_by_user_id`, `updated_by_user_id`, `version_serial` — are injected by the loader automatically. The developer does not declare them. The versioning sibling table (`task_version`) is generated automatically when `versioned: true` is set.

3.2 Type Selection The nine types serve specific purposes.

int — whole numbers with declared bounds. Use for: counts, quantities, priorities, ratings, sequence numbers, years. Always declare min and max. Prefer int over float for anything that should not have fractional values — currency in minor units (cents, not dollars), quantities of discrete items, scores on fixed scales.

float — decimal numbers with declared bounds. Use for: measurements, percentages, coordinates, rates. Always declare min and max. Be aware of floating-point representation — for financial calculations where precision matters absolutely, use int in minor units and let runners handle display formatting.

varchar — bounded-length strings. Use for: names, titles, identifiers, short descriptions, codes, slugs. Always declare length. The length is a real constraint — the database enforces it. Choose lengths deliberately: 63 for identifiers (DNS-compatible), 255 for titles and names, 1024 for short descriptions or URLs.

text — unbounded strings. Use for: long-form content, descriptions, notes, markdown content, prose. No length constraint. Use sparingly — most string fields have a natural upper bound and should be varchar.

boolean — true or false. Use with the naming conventions: `is_` prefix for present state (`is_active`, `is_published`, `is_billable`), `was_` prefix for past events (`was_approved`, `was_escalated`, `was_migrated`).

datetime — timestamp with timezone. Use for: event times, deadlines, schedule times. Always suffix with `_time` (`created_time`, `approved_time`, `expires_time`). Stored in UTC. Timezone handling is a runner concern, not a schema concern.

date — calendar date without time. Use for: due dates, birth dates, effective dates, billing cycle dates. Always suffix with `_date`. Use date instead of datetime when the time component is meaningless — a task due date is a date, not a datetime.

json — structured JSON with discriminator-based validation. Use for: typed payloads where different subtypes have different field sets (runner specs, schedule payloads, monitor configurations, evidence records). Always pair with a discriminator enum field — the `*_type` field determines which JSON schema validates the `*_data_json` field. Do not use json as a catch-all for unstructured data unless building the graduated formalization pattern (generic collection → proper schema).

enum — constrained value set. Use for: status fields, type discriminators, classification levels, category tags. Declare all values explicitly. Remember: enum values can be added but never removed (schema evolution rule). Plan the initial value set to cover known cases and leave conceptual room for additions.

foreign_key — reference to another entity. Use for: ownership, membership, reference, hierarchy. Always name as `referenced_table_id` or `role_referenced_table_id` when multiple FKs reference the same target.

3.3 Constraint Selection Six constraint mechanisms are available.

Numeric range (min/max on int and float). Enforced on every write. Choose bounds that reflect domain reality — priority 1–5, percentage 0.0–100.0, age 0–200, price 0–999999999 (in minor units). Bounds can be widened but never narrowed (evolution rule), so start with bounds that are correct, not bounds that are generous.

String length (length on varchar). Enforced on every write. Choose lengths that match the domain — 63 for machine identifiers, 255 for human-readable names, 1024 for descriptions. Lengths can be widened but never narrowed.

Enum value set (values on enum). Enforced on every write. Values can be added but never removed. The existing value “pending” exists forever even if deprecated. Plan initial sets carefully.

Foreign key reference (references on foreign_key). Enforced on every write — the referenced row must exist. This is relational integrity maintained by the database engine, not application code.

Nullable (nullable modifier). When false, the field must have a value on every write. When true, the field may be null. Default is false — fields are required unless explicitly marked nullable. This is a deliberate default: most fields should have values, and nullable fields should be the exception that the developer opts into.

Unique (unique modifier). When true, no two active rows in the table may have the same value for this field. Uniqueness applies to the active population — soft-deleted rows (`is_active=false`) do not participate in uniqueness checks. Uniqueness can be added to a field on an empty table but cannot be added to a field with existing duplicates, and cannot be tightened on a populated field (evolution rule).

3.4 The Discriminator Pattern When entities of the same structural type have different typed payloads — different kinds of monitors, different kinds of schedules, different kinds of evidence records — the discriminator pattern applies.

```
name: monitor
domain: monitoring
versioned: true
fields:
  - name: name
```

```

type: varchar

length: 255

nullable: false

  - name: monitor_type

type: enum

values: [prometheus_query, http_probe, tcp_probe, script_local,
        script_remote, k8s_event_watch, cloud_metric]

nullable: false

  - name: monitor_data_json

type: json

nullable: false

  - name: is_enabled

type: boolean

nullable: false

default: true

```

The `monitor_type` enum selects which JSON schema validates `monitor_data_json`. Each JSON schema is a separate YAML file under `schema/json_schemas/monitor/`. The loader registers the mapping. The gate pipeline validates the JSON payload against the schema selected by the discriminator value on every write.

This pattern replaces inheritance. There is no base class with subclasses. There is one table with a discriminator field and a validated JSON payload. All monitors share the same structural fields (`name`, `type`, `is_enabled`). Each monitor kind has its own validated payload shape. Adding a new monitor kind means adding an enum value and a JSON schema file — no code change.

3.5 Bridge Table Design Bridge tables handle many-to-many and polymorphic relationships with clean foreign key integrity.

Many-to-many bridge:

```

name: task_label

domain: project_management

versioned: false

fields:
  - name: task_id
type: foreign_key
references: task
nullable: false

  - name: label_id
type: foreign_key
references: label
nullable: false

  - name: applied_by_user_id
type: foreign_key
references: ops_user
nullable: false

```

The bridge table has no identity beyond its foreign keys. It exists to relate two entities. It may carry additional fields — who applied the label, when, why — that describe the relationship, not the related entities.

Polymorphic bridge:

When an entity (comment) can be attached to multiple different entity types (task, project, milestone), create one bridge table per target type:

```

name: task_comment

domain: project_management

versioned: false

```

```

fields:
    - name: task_id
type: foreign_key
references: task
nullable: false

    - name: comment_id
type: foreign_key
references: comment
nullable: false

name: project_comment
domain: project_management
versioned: false

fields:
    - name: project_id
type: foreign_key
references: project
nullable: false

    - name: comment_id
type: foreign_key
references: comment
nullable: false

```

This is more tables than a single polymorphic FK column, but every foreign key is a real foreign key to a real table, checked by the database on every write. No type column selecting which table the FK points to. No orphaned references. No application-level

integrity checking. The database enforces it mechanically.

3.6 Hierarchy Design Self-referential foreign keys model hierarchies — organizational units, task/subtask trees, location trees, category trees.

```
name: category
domain: catalog
versioned: true
fields:
  - name: name
type: varchar
length: 255
nullable: false
  - name: parent_category_id
type: foreign_key
references: category
nullable: true
  - name: depth
type: int
min: 0
max: 20
nullable: false
default: 0
```

The `parent_category_id` is nullable — root categories have no parent. The `depth` field is maintained by a runner that traverses the parent chain and sets depth on create or reparent. The depth bound (max: 20) prevents unbounded hierarchy depth.

Traversal uses `get_dependencies` with the parent chain join path. The search

API can filter by parent, by depth, or by subtree membership. A runner that computes hierarchy metadata — full path string, child count, leaf status — writes these as fields on the entity or as observation cache rows, depending on whether the computed values are governed state or derived cache.

3.7 Governance Field Configuration Three governance fields control per-entity access beyond role-based authorization.

****_requires_group**** scopes entity visibility to members of a specific group. When set, only members of the referenced group (or administrators) can read or write the entity. Use for: project-scoped data (only project members see project tasks), team-scoped data (only team members see team resources), personal data (only the owner sees their private items).

```
- name: _requires_group
type: foreign_key
references: ops_group
nullable: true
```

When null, the entity is visible to any authenticated user with appropriate role permissions. When set, group membership is checked at gate step 2 layer 2.

****_access_classification**** controls field-level visibility. When a field has an access classification, only users with roles that include that classification level can see the field value. Use for: salary data (restricted to HR), medical details (restricted to care team), financial details (restricted to billing administrators).

The classification is set per entity row, not per field definition. This means different rows of the same entity type can have different classification levels. A patient record for a VIP might have higher classification than a standard record.

versioned: true opts the entity into change management. Every write produces a version row. Change sets route through approval. Version history is queryable. This is the default for entities where change history matters. Set to false for ephemeral or high-frequency entities where versioning would create noise — observation cache tables, job records, temporary coordination state.

3.8 Draft Mode Configuration For entities where interactive editing is the primary use pattern, draft mode governance flags reduce friction without eliminating structural safety.

```
name: document
domain: knowledge_management
```

```
versioned: true
```

```
draft_mode:
```

```
  _autoversion_disabled: true
```

```
  _edit_latest_version: true
```

```
  _audit_logs_disabled: true
```

```
notes: |
```

```
Draft mode enabled. Interim editing states are not individually  
versioned or audited. Version commits engage full governance.
```

```
Properties weakened: per-write versioning, per-write audit.
```

```
Properties preserved: schema validation, authorization,  
committed-version history, committed-version audit.
```

```
Acceptable because: document content is iterative drafting  
where per-keystroke versioning creates noise without value.
```

The property analysis in the notes field is not optional guidance — it is the schema steward's verification that the tradeoff is understood. Before setting draft mode flags, the developer must enumerate which properties are weakened and confirm that the weakened properties are not required for the entity's role in the application.

Auth and authorization always run. Schema validation always runs. Bound validation always runs. Draft mode never weakens security or structural integrity. It weakens recording — versioning, audit, change management — for interim saves between explicit commits.

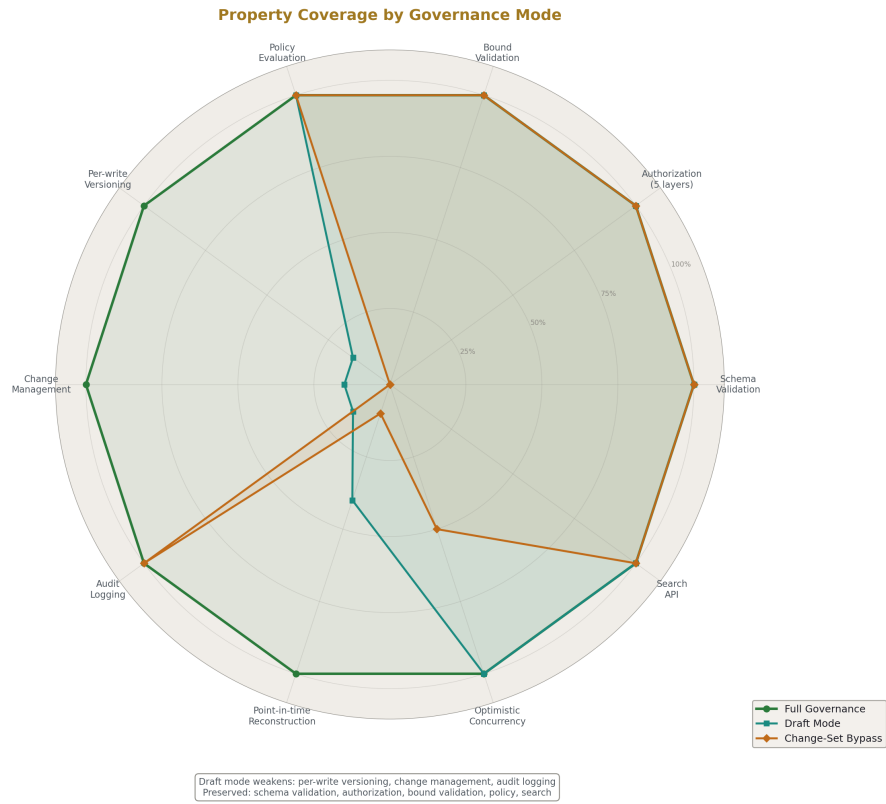


Figure 2: Fig. 8: Property Impact Radar — full governance covers all 10 axes, draft mode contracts versioning/audit/change-mgmt while holding validation and authorization.

3.9 Schema Organization Schema files are organized in a directory structure that mirrors the domain taxonomy.

```

schema/

  directory.yaml          # imports all domain directories

  conventions/

reserved.yaml            # reserved field definitions

domains/

01_identity/             # users, groups, roles

```

```

02_core/                # primary domain entities
03_workflow/            # lifecycle and process entities
04_policy/              # policies, rules, classifications
05_schedule/            # schedules, deadlines
06_integration/         # authority pointers, external refs
07_observation/         # observation cache tables
08_audit/               # audit, evidence, compliance
09_runner/              # runner specs, jobs, capabilities
    json_schemas/
monitor/                # per-discriminator JSON schemas
schedule/
evidence_record/
runner_spec/
    meta/
_schema_meta.yaml      # schema metadata entity types

```

The directory numbering creates a natural dependency order — identity entities exist before core entities reference them, core entities exist before policies reference them. The `directory.yaml` file imports domain directories in order. Entity files are leaf-level — they do not import other files.

4. Policy Construction Methods

Policies govern application behavior through data rows evaluated by the gate pipeline. Every policy is itself an OpsDB entity — versioned, change-managed, and auditable. Changing application behavior means submitting a change set that modifies policy rows, not deploying new code.

4.1 Approval Rule Design Approval rules determine which changes require human approval and from whom. An approval rule is a policy row that matches against change

set characteristics and declares the approval requirements.

The matching criteria include: entity type (which tables are affected), field names (which specific fields are changing), namespace (which organizational scope), data classification level (how sensitive the data is), security zone (which environment), and the proposer's role (different rules for human vs automated proposers).

Design approval rules from the stakes, not from the entity structure. Ask: if this change is wrong, what is the impact? A task status change to “done” has low impact — auto-approve. A project budget change has financial impact — require the project owner's approval. A data classification change affects who can see what — require the data steward's approval. A production policy change affects system behavior — require two approvers from different roles (segregation of duties).

The calibration for AI proposers follows the same logic. An AI runner that proposes changes to low-sensitivity observation summaries can be auto-approved. An AI runner that proposes changes to financial records based on reconciliation results requires human review. The policy rows that govern AI-proposed changes are the same mechanism as all other approval rules — they just match on the proposer's service account identity.

```
# Approval rule: budget changes require project owner

name: budget_change_approval

policy_type: approval_rule

policy_data_json:

  match:

entity_types: [project]

field_names: [budget_amount, budget_currency]

  require:

approver_count: 1

approver_source: ownership_bridge

approver_role: project_owner

# Approval rule: AI-proposed entity changes require human review

name: ai_proposal_review
```

```
policy_type: approval_rule

policy_data_json:

    match:

proposer_type: service_account

proposer_role: ai_runner

    require:

approver_count: 1

approver_source: entity_owner
```

4.2 Access Control Design Access control uses three complementary mechanisms.

Role-based access (layer 1). Define roles that map to job functions: admin, manager, member, viewer, billing_admin, hr_admin. Assign permissions per role: which entity types can be read, which can be written. This is the coarsest control — it determines what categories of data a role can interact with.

Entity-level scoping (layer 2). Set `_requires_group` on entities to restrict visibility to group members. A project’s tasks are visible only to project members by setting `_requires_group` to the project’s member group. This scoping is per-row — different projects have different groups, so different teams see different projects.

Field-level classification (layer 3). Set `_access_classification` on entity rows to control which fields are visible. A patient record with classification “restricted” hides diagnosis and treatment fields from roles that lack the restricted access grant. The API response includes the field names with a metadata flag indicating the value is hidden, so the frontend can render “access restricted” rather than an empty field.

Design access control from the data sensitivity outward, not from the user roles inward. Ask: what data exists, how sensitive is each element, who legitimately needs to see it? Then create the roles and groups that express those access needs. This prevents the common failure mode of designing roles first and discovering later that the role structure doesn’t match the data sensitivity boundaries.

4.3 State Transition Rules For entities with lifecycle state machines, transition rules are policy rows that the gate pipeline evaluates at step 5.

```
name: task_transitions

policy_type: state_transition
```

```

policy_data_json:
    entity_type: task
    field: status
    transitions:
- from: draft
    to: [open]
- from: open
    to: [in_progress, draft]
- from: in_progress
    to: [review, draft]
- from: review
    to: [done, in_progress]
- from: done
    to: [archived]
- from: archived
    to: []      # terminal state

```

When a change set proposes a status field change on a task, the gate pipeline checks the transition against this policy. If the transition is not in the declared set, the change set is rejected. The rejection is audited — the audit log records the attempted invalid transition, who proposed it, and when.

Transition rules can include conditions evaluated by runners. “A task can only move to done if all subtasks are done” is not expressible as a static policy row — it requires querying subtask state. A verifier runner handles this: before a task can transition to done, the runner checks subtask completion and writes an evidence record. The approval rule for the done transition can require this evidence as a precondition.

4.4 Retention Policy Design Retention policies determine how long versioned data, observation cache entries, and audit records persist.

```

name: task_retention

policy_type: retention

policy_data_json:

    entity_type: task

    version_history_days: 365

    observation_cache_days: 30

    audit_log_days: 2555      # 7 years for compliance

```

A reaper runner reads retention policies and enforces them — soft-deleting version rows past their retention horizon, removing expired observation cache entries, preserving audit entries for their declared retention period.

Design retention from regulatory requirements first, operational needs second, storage costs third. Compliance-relevant entities may have retention periods set by law (7 years for financial records, varies by jurisdiction for medical records). Operational entities use retention periods that match their operational relevance — 30 days for observation cache, 90 days for runner job records. Storage cost should never drive retention for compliance-relevant data.

4.5 Cross-Field Invariant Rules The schema vocabulary does not include conditional constraints. Cross-field invariants — “if status is active then start_date must be set,” “if payment_method is credit_card then card_token must be present” — are policy rows evaluated at gate step 5.

```

name: active_requires_start_date

policy_type: semantic_invariant

policy_data_json:

    entity_type: task

    condition:

        field: status

        equals: active

        require:

```

```
field: start_date

not_null: true

message: "Active tasks must have a start date"
```

This separation keeps the schema vocabulary simple and the validation logic inspectable. Every cross-field rule is a queryable policy row. You can enumerate all invariants for an entity type with a single search query. Changing an invariant is a change set, audited and reversible. In conventional development, cross-field validation is buried in application code — scattered across model validators, service methods, and middleware — and discovering all the rules requires reading the codebase.

5. Runner Construction Methods

Runners implement the backend logic that declarations cannot express. Each runner follows the get-act-set pattern, uses the shared library suite for infrastructure concerns, and operates within declared bounds.

5.1 Runner Design Method Before writing code, design the runner by answering seven questions.

What does it read? List every OpsDB entity type and external source the runner consults. A notification runner reads: change set entities (to detect new approvals needed), schedule entities (to detect approaching deadlines), on-call assignment entities (to resolve recipients), authority entities (to get notification channel configuration).

What does it write? List every OpsDB table and external system the runner modifies. The notification runner writes: observation cache entries (delivery receipts), and dispatches to external notification channels (email, webhook, chat) via the library suite.

What is the gating mode? Direct write for observation-like output that bypasses change sets but is still authenticated, validated, and audited. Auto-approve for low-risk changes that create change sets but transition through approval automatically. Approval-required for high-risk changes that route to human reviewers.

What triggers execution? Scheduled (cron or rate-based — runs every N minutes), event-driven (triggered by a state change the runner watches for), or long-running (continuous loop with sleep between cycles).

What are the bounds? Retry budget (how many times to retry a failed action), execution time limit (maximum seconds per cycle), scope per cycle (maximum entities to process per run), memory bounds (maximum heap allocation). These are declared in the runner spec and enforced by the library suite.

What does idempotency mean? If the runner runs twice on the same input, what happens? For a notification runner: sending the same notification twice is undesirable, so the

runner tracks which notifications have been sent using observation cache rows keyed by change set ID and recipient. For a reconciler: applying the same correction twice produces the same end state because the change set is idempotent — if the value is already correct, no change is proposed.

What report keys can it write? The runner declares which observation cache tables and fields it can write to. The API and library suite both validate writes against these declarations. A notification runner can write to `observation_cache_state` with keys related to notification delivery. It cannot write to `observation_cache_metric` or to governed entity tables outside its declared scope.

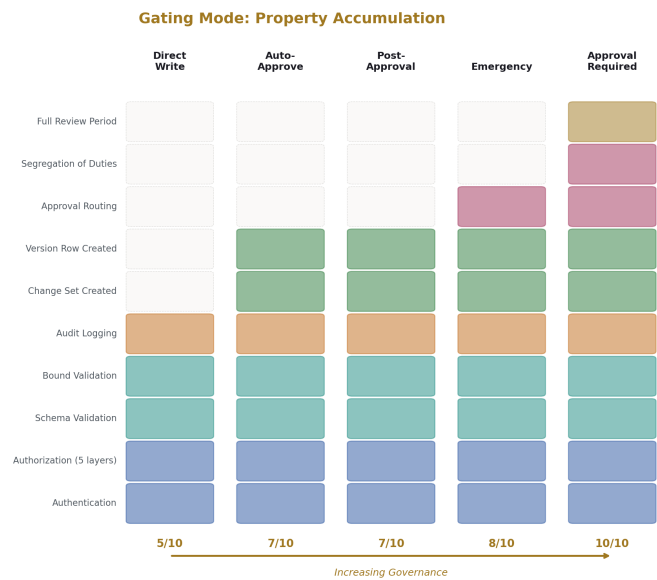


Figure 3: Fig. 3: Gating Mode Property Accumulation — each mode adds governance properties, from 5/10 (direct write) to 10/10 (approval-required).

5.2 Puller Runner Construction A puller runner imports data from an external system into OpsDB observation cache tables.

Structure:

```
get phase:

- read runner spec from OpsDB (connection params, schedule, bounds)

- read authority entity from OpsDB (endpoint URL, auth reference)
```

```

- read last sync cursor from observation cache (if incremental)

act phase:

- connect to external API via library suite

- paginate through external data

- transform each external record to OpsDB entity shape

- handle rate limits (library suite manages backoff)

set phase:

- write transformed records to observation cache via API

- update sync cursor in observation cache

- write sync summary (record count, errors, duration) to observation cache

```

Schema mapping. The transformation from external format to OpsDB entity shape is the puller’s primary domain logic. A Stripe payment puller reads Stripe’s payment intent objects and maps them to OpsDB observation cache rows: `stripe_payment_id` → observation key, `amount` → `observed_amount`, `status` → `observed_status`, `created` → `observed_time`. The mapping is explicit code in the runner — typically a function per external entity type, 20–40 lines each.

Incremental sync. Most external APIs support cursor-based pagination or timestamp-based filtering. The puller stores its last sync cursor as an observation cache entry. Each cycle reads from the cursor forward. Full resync is triggered by deleting the cursor entry — the puller starts from the beginning on its next cycle. This is idempotent: running a full resync produces the same end state as incremental sync that never missed an event.

Freshness. Every observation cache entry written by a puller carries a freshness timestamp — the time the puller read the data from the external source. Consumers of the observation data can filter by freshness — “show me only observations from the last hour” — to avoid acting on stale data. The search API’s freshness annotation parameter makes this a standard query predicate.

5.3 Reconciler Runner Construction A reconciler compares desired state (from governed entities) with observed state (from observation cache or external queries) and proposes or applies corrections.

Structure:

```

get phase:

```

- read desired state from governed entities via search API
- read observed state from observation cache via search API
- (or read observed state directly from external system via library suite)

act phase:

- for each desired entity, find the corresponding observed state
- compute the diff: fields where desired $\neq$ observed
- for each diff, determine the correction action
- filter out diffs within acceptable tolerance

set phase:

- for each correction:
- if auto-approve: submit change set with correction, auto-approved by policy
- if approval-required: submit change set with correction, routes to approvers
- if direct external action: execute via library suite, write result as observed

Convergence behavior. The reconciler is level-triggered, not edge-triggered. It does not react to events (“this value changed”). It compares current desired state with current observed state on every cycle. If an event is missed — a webhook fails, a notification drops — the next reconciler cycle detects the discrepancy from the state comparison. This is the R3 idempotency principle and R8 level-triggered-over-edge-triggered from the ops book applied to runner design.

Tolerance. Not every diff is worth correcting. A reconciler comparing expected inventory counts with warehouse-observed counts might tolerate a variance of ± 2 items before proposing a correction. Tolerance thresholds are configurable in the runner spec’s typed JSON payload — they are data, not hardcoded constants.

Partial drift. When the reconciler finds multiple diffs across multiple entities, it can propose a single bulk change set covering all corrections, or individual change sets per entity. Bulk change sets are atomic — all corrections apply or none do. Individual change sets allow partial progress — some corrections can be approved while others are pending review. The choice depends on whether the corrections are interdependent. Independent corrections (each entity’s count is independently correct or incorrect) use individual change sets. Interdependent corrections (a set of accounts must balance to zero) use bulk change sets.

5.4 Verifier Runner Construction A verifier checks that expected conditions hold and produces evidence records documenting the result.

Structure:

get phase:

- read schedules that are due for verification
- read target entities referenced by each schedule
- read prior evidence records (to detect regressions)

act phase:

- for each due schedule:
- evaluate the verification condition against the target
- determine pass or fail
- assemble supporting data (what was checked, what was found)

set phase:

- write evidence record per verification with:
- pass/fail result
- verification time
- supporting data as typed JSON payload
- reference to schedule and target entity
- update schedule's last_verified_time

Evidence record typed payloads. Each verification type has its own JSON schema for the evidence record payload. A certificate validity verifier writes: certificate subject, issuer, expiration date, days until expiration, chain validation result. A backup verification verifier writes: backup timestamp, backup size, restore test result, duration. The typed payload ensures every evidence record of a given type has consistent structure — queryable, comparable across time, and usable by compliance reporting.

Regression detection. The verifier reads prior evidence records for the same target. If the prior result was pass and the current result is fail, the verifier can trigger escalation — writing an alert entity, notifying the on-call team, or proposing a change set that flags the

entity for remediation. Regression detection is domain logic in the runner, not infrastructure — the runner decides what constitutes a regression for its verification type.

5.5 Notification Runner Construction A notification runner detects state transitions that warrant notification and dispatches messages through configured channels.

Structure:

get phase:

- read recent change sets (since last notification cycle)
- read pending approval requests
- read approaching deadlines (schedules with due dates within threshold)
- read notification configuration (which transitions trigger which channels)
- read recipient resolution data (on-call assignments, ownership bridges)

act phase:

- for each trigger event:
 - resolve recipients from ownership/stakeholder bridges and on-call assignments
 - select channel(s) from notification configuration
 - format message using configured template
 - deduplicate against prior notifications (check observation cache)

set phase:

- dispatch via library suite (email, webhook, chat API)
- write delivery receipt to observation cache (keyed by event + recipient)
- write delivery failures to observation cache for retry on next cycle

Channel configuration as data. Notification channels are authority entities with typed payloads: email (SMTP configuration), webhook (URL, headers, auth), chat (Slack/Teams API configuration). Adding a notification channel means creating an authority entity through a change set. Removing a channel means deactivating the authority entity. The runner reads the active authority entities and dispatches through

them. No code change for channel changes.

Recipient resolution. The runner does not maintain its own recipient lists. It resolves recipients from OpsDB data: ownership bridges (who owns the affected entity), stakeholder bridges (who has declared interest), on-call assignments (who is currently responsible), and escalation paths (who to contact if the primary recipient does not respond within a configured window). All of these are governed entities — changing who receives notifications means changing data, not code.

5.6 Configuration Push Runner Construction A configuration push runner reads governed state from OpsDB and pushes it to an external system in that system’s native format.

Structure:

`get phase:`

- read governed entities that the external system consumes
- read the external system's current configuration (via library suite)
- read the runner spec for transformation rules and connection params

`act phase:`

- transform governed entities into the external system's native format
- compute the diff between current external config and desired config
- generate the update payload

`set phase:`

- push the update to the external system via library suite
- write push result (success/failure, timestamp, diff summary) to observatio
- on failure: log structured error, increment retry counter, respect retry bu

Transformation as domain logic. The transformation from OpsDB entity shape to external format is the runner’s primary contribution. An API gateway configuration push runner reads route definition entities, rate limit policy entities, and consumer quota entities from OpsDB, and produces the gateway’s native configuration format — a YAML file, a JSON API payload, a protobuf message. This transformation is 50–100 lines of mapping code. The library suite handles the connection, authentication, retry, and error handling for the push.

Diff-based push. The runner does not blindly push the full configuration on every cycle. It reads the external system's current configuration, computes the diff against the desired configuration from OpsDB, and pushes only the changes. This minimizes disruption to the external system and makes each push operation smaller and faster. If the diff is empty — the external system already matches the governed state — the runner does nothing and writes a “no changes” observation.

Cache independence. The external system should cache the configuration it receives. If OpsDB is unavailable, the external system continues operating with its cached configuration. When OpsDB returns, the runner catches up and pushes any accumulated changes. The external system never depends on OpsDB at runtime — only at configuration refresh time. This is the local cache plus global truth principle from the ops book: the external system has a local copy, OpsDB is the global truth.

5.7 Observation Pull Runner Construction An observation pull runner reads results from an external system and writes them to OpsDB observation cache tables.

This is the complement of the configuration push runner. Together they form the bridge pattern for split-backend and wrapper architectures: push configuration out, pull results back.

Structure:

get phase:

- read runner spec (external system connection, polling interval, scope)
- read authority entity (endpoint, auth reference)
- read last poll cursor from observation cache

act phase:

- connect to external system's results API via library suite
- read results since last poll cursor
- transform results to observation cache format

set phase:

- write results to observation cache with freshness timestamps
- update poll cursor
- write poll summary (record count, errors, latency) to observation cache

The pull runner handles the external system's eventual consistency. If the external system reports results with delay — a payment processor that settles transactions hours after authorization — the pull runner writes observations with the timestamp from the external system, not the pull time. Consumers querying the observation cache see when the event actually occurred, not when OpsDB learned about it.

5.8 AI Observation Runner Construction An AI observation runner uses an LLM to generate derived observations over governed data. The output is always observation cache — never governed state.

Structure:

get phase:

- read runner spec (model, token budget, prompt template reference, regeneration trigger, target entity types)
 - read governed entities that form the context
 - traverse relationships to build complete context
- (entity → dependencies → change sets → alerts → observations)

act phase:

- assemble context into structured prompt
- call LLM via library suite (with retry, timeout, circuit breaking)
- validate response length against configured character budget
- validate response structure if structured output is expected

set phase:

- write AI observation to observation cache with:
 - freshness timestamp
 - character budget used
 - model identifier
 - source entity references (which entities were summarized)

- prompt hash (for reproducibility tracking)
- on LLM failure: write error observation, continue to next entity

Configuration via runner spec:

```

name: incident_summary_runner
runner_type: ai_observation
runner_data_json:
  model: "claude-sonnet"
  max_tokens: 500
  character_budget: 2000
  regeneration_trigger: "source_entity_change"
  target_entity_types: [incident]
  context_traversal:
    - incident -> affected_service (via bridge)
    - affected_service -> service_connection (dependencies)
    - incident -> change_set (recent, last 24h)
    - incident -> alert_fire (related alerts)
  prompt_template_ref: "incident_summary_v1"
  observation_cache_table: observation_cache_state
  observation_key_prefix: "ai_summary_incident_"

```

Multiple granularities. The same runner code serves different summary needs through different runner specs. An 80-character headline for a dashboard ticker has `character_budget: 80` and a terse prompt template. A 2000-character incident summary for the war room has `character_budget: 2000` and a detailed prompt template. A 12-character severity classification has `character_budget: 12` and a classification-focused prompt template. Three runner specs, one runner implementation.

Regeneration. The runner regenerates observations when source data changes. The

`regeneration_trigger` field in the spec determines when: on any source entity change, on a schedule, or on demand. On-change regeneration keeps summaries fresh as incidents evolve. Scheduled regeneration limits LLM costs for stable data. On-demand regeneration lets operators request fresh summaries when needed.

Containment. If the LLM hallucinates, the governed data is unaffected. The AI observation is a cache row that will be overwritten on the next regeneration cycle. The freshness timestamp tells consumers how recent the summary is. The source entity references let consumers verify the summary against the actual data. The prompt hash enables reproducibility investigation — given the same context and prompt, does the LLM produce the same hallucination?

5.9 Custom Domain Runners Some domain logic does not fit the standard runner kinds. A grade computation runner, an ELO rating runner, a budget tracking runner, an invoice generation runner — each performs domain-specific computation that reads governed state and writes results back.

The get-act-set pattern still applies. The library suite still handles infrastructure. The runner still declares its scope and bounds.

Example: invoice generation runner.

get phase:

- read active subscriptions with `billing_cycle_date` $\leq$ today
- read plan entities for each subscription (pricing data)
- read usage records since last invoice for metered plans
- read tax policy data for jurisdiction-based tax rates
- read credit balance for each customer account

act phase:

- for each billable subscription:
- compute base charge from plan pricing
- compute usage charges from usage records $\times$ metered rates
- compute proration for mid-cycle changes
- apply credits from credit balance

- compute tax from jurisdiction rules
 - generate line items
- set phase:
- submit change set creating invoice entity with line items
 - submit change set updating credit balance if credits applied
 - write invoice generation summary to observation cache

The runner is 200–300 lines. The pricing data, tax rules, proration logic, and credit application are domain computations — the runner’s actual contribution. Everything else — reading data, writing change sets, handling errors, retrying failures, logging, authentication — is library suite calls.

5.10 The Server Runner Model Runners operate as threads within a server runner process. The server runner is a single Go binary that hosts multiple runner threads, manages their lifecycle, and shares infrastructure resources.

Thread configuration. The server runner reads runner spec entities from OpsDB to determine which runners to instantiate and how many threads each gets. Configuration is data — changing which runners run, how many threads each gets, and what cycle timing each uses means changing runner spec rows through change sets.

```
# Runner thread configuration (in runner_spec entity)

name: notification_runner_spec

runner_type: notification

runner_data_json:

  thread_count: 2

  cycle_interval_seconds: 30

  retry_budget: 3

  execution_timeout_seconds: 60

  scope_per_cycle: 100
```

Shared resources. All runner threads share the database connection pool, the library suite instances (with per-runner scope enforcement), the logging infrastructure, and the

metrics collection. This eliminates the per-runner operational overhead that separate processes would require — no per-runner deployment, no per-runner monitoring configuration, no per-runner resource allocation.

Scope isolation. Despite sharing a process, runner threads are isolated by scope. Each thread's library suite instance is configured with the runner's declared capabilities and targets. A notification runner thread cannot write to billing tables because its scope declaration does not include them. The library suite checks every API call against the thread's scope before sending it. The API gate checks again on receipt. Double enforcement — library-side and API-side — means a bug in one runner thread cannot affect data outside its declared scope.

Scaling. When the server runner needs more capacity, you run additional instances of the same server runner binary. Each instance reads the same runner specs and distributes work through the standard coordination mechanisms — advisory locks, runner job claims, and observation cache cursors. The 0/1/N rule applies: one server runner type, N instances for scaling.

Health monitoring. Each runner thread writes health observations to the observation cache on every cycle: cycle start time, cycle duration, records processed, errors encountered. A monitoring runner (or an external monitoring system pulling from the observation cache) detects unhealthy threads — prolonged cycle times, increasing error counts, stuck threads. The health observations are the same mechanism as any other observation — queryable through the search API, subject to freshness semantics, and visible to dashboards.

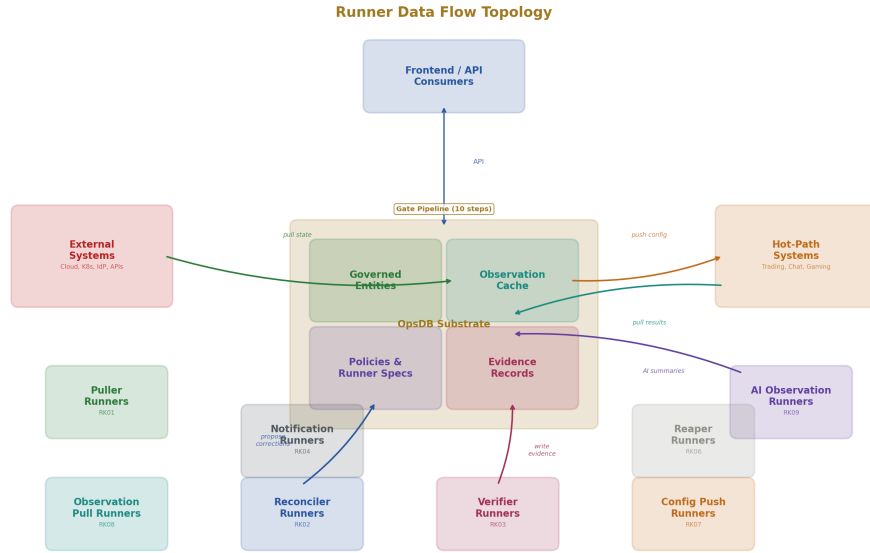


Figure 4: Fig. 5: Runner Data Flow — pullers bring external state in, reconcilers compare, verifiers produce evidence, config push sends governed state out.

6. Frontend Integration Methods

The frontend is a thin consumer of the OpsDB API. It has no database. It implements no validation, authorization, versioning, or audit logic. It authenticates users, translates user actions into API calls, and renders responses.

6.1 Authentication Integration The frontend authenticates users through SSO delegation to an identity provider. The OpsDB API accepts tokens from the configured IdP(s). The frontend’s responsibility is acquiring the token (through OAuth2 / OIDC flows) and passing it with each API request.

The frontend does not maintain user sessions beyond the token lifecycle. Session management is token management — refresh tokens, handle expiry, redirect to IdP for re-authentication. The OpsDB API validates the token on every request at gate step 1. No session state in the frontend’s backend-for-frontend layer. No session database. No session synchronization across frontend instances.

6.2 Read Patterns **List views.** Call `search` with filter predicates, projection (which fields to return), ordering, and cursor-based pagination. The search API returns only entities the authenticated caller can access — multi-tenancy is invisible to the frontend code.

Fields the caller cannot see due to access classification are omitted from the response with metadata indicating the omission.

```
POST /api/search
```

```
{
  "entity_type": "task",
  "filters": {
    "and": [
      {"field": "project_id", "op": "eq", "value": "proj_123"},
      {"field": "status", "op": "in", "value": ["open", "in_progress"]},
      {"field": "assignee_user_id", "op": "eq", "value": "current_user"}
    ]
  },
  "fields": ["id", "title", "status", "priority", "due_date", "assignee_user_"],
  "order_by": [{"field": "priority", "direction": "asc"}, {"field": "due_date", "direction": "desc"}],
  "page_size": 25,
  "cursor": null
}
```

Detail views. Call `get_entity` with the entity type and ID. The response includes all fields the caller can see plus metadata: version serial, created/updated timestamps, created/updated by identity. If the entity is versioned, the current version is returned. Named join paths can include related entities in a single request — a task detail view can include the project, the assignee, and recent comments through declared join paths.

Activity feeds. Call `get_entity_history` to retrieve the version chain for an entity. Each version row includes the full state at that version, the change set that produced it, the proposer, the approver(s), and the timestamp. The frontend renders a timeline of changes with attribution. For a project-level activity feed, query change sets filtered by entity types within the project's scope.

Dashboard and reporting. Dashboards are search queries with aggregation handled by the frontend or by observation cache tables populated by reporting runners. Simple counts

and distributions can be computed client-side from search results. Complex aggregations — time-series trends, cross-entity correlations, statistical summaries — are computed by a reporting runner that writes results to observation cache tables, which the frontend reads through the same search API.

Point-in-time views. Call `get_entity_at_time` with a timestamp to reconstruct the state of any entity at any prior point. The response is the version row that was current at the specified time — a single row lookup, not a chain replay. This supports audit views (“what did this record look like on March 15th”), incident investigation (“what was the configuration when the outage started”), and regulatory queries (“show the state of this account at end of fiscal year”).

6.3 Write Patterns **Standard writes.** The frontend submits a change set containing the proposed field changes and a reason string.

```
POST /api/change_set/submit

{
  "entity_type": "task",
  "entity_id": "task_456",
  "changes": [
    {"field": "status", "old_value": "open", "new_value": "in_progress"},
    {"field": "assignee_user_id", "old_value": null, "new_value": "user_789"}
  ],
  "reason": "Starting work on authentication redesign task"
}
```

The `old_value` fields enable optimistic concurrency — if the entity has changed since the frontend read it, the submit fails with a conflict response. The frontend can then re-read, merge if appropriate, and re-submit.

Auto-approved writes. For changes matching auto-approve policy rules, the change set transitions through draft → submitted → approved → applied within the same request cycle. The frontend experiences this as an immediate write. But the change set is created, the version row is written, the audit entry is logged, and the change is reversible through a subsequent change set.

Approval-required writes. For changes matching approval-required policy rules, the change set transitions to pending_approval. The API response includes the required ap-

provals — who needs to approve, from what roles, how many approvals are needed. The frontend renders a pending state: “Your change to project budget is awaiting approval from the project owner.”

The frontend for approvers shows pending change sets with the proposed changes, the reason, the proposer’s identity, and the approval/rejection controls. Approving or rejecting is itself an API call — `approve_change_set` or `reject_change_set` with a reason. When all required approvals are received, the change set executor runner applies the changes.

Entity creation. Creating a new entity is a change set where all fields are `new_value` with no `old_value`. The change set creates the entity, the version row, and the audit entry atomically.

Bulk operations. The `bulk_submit` operation accepts multiple entity changes in a single change set. The changes are applied atomically — all succeed or all fail. Use for operations that must be consistent across entities: creating an invoice with line items, assigning multiple tasks to a sprint, applying a configuration change across multiple entities.

Emergency path. For break-glass situations, the `emergency_apply` operation bypasses normal approval routing. It requires the caller to have the emergency role. It applies the change immediately with reduced approvals. It sets a mandatory flag in the audit log. It creates a review schedule entry requiring post-incident review within a configurable window. The frontend for emergency operations should be distinct from the normal editing interface — a separate UI path that makes the extraordinary nature of the action visible.

6.4 Draft Mode Integration For entities with draft mode governance flags enabled, the frontend implements continuous autosave with explicit version commits.

Autosave. On every user edit (debounced — typically 1–3 seconds after the last keystroke), the frontend sends a direct write to the API. Auth, schema validation, and bound validation run. Versioning, change management, and audit logging are skipped per the draft mode flags. The current row is updated in place. If the browser crashes, the user loses at most a few seconds of work.

Version commit. When the user explicitly saves (a deliberate action — a button click, a keyboard shortcut, not an autosave), the frontend sends a version commit request. All ten gate steps run. A version row is created with the full state. A change set records the delta from the prior committed version. The audit log records the commit event. The commit becomes part of the permanent version history.

Rendering draft vs committed state. The frontend distinguishes between the current working state (the entity’s current row, which may include uncommitted edits) and the committed version history (the version rows). An editing interface shows the current working state with a “save version” button. A history interface shows committed versions with diffs between them. The working state is ephemeral — it represents the user’s current draft. The committed versions are permanent — they represent intentional checkpoints.

Undo/redo. Undo within a draft session is a frontend concern — standard browser undo or application-level undo stack. Undo across committed versions is a governed operation — the frontend submits a change set that restores the prior version’s field values. This change set goes through full governance: validation, authorization, approval (if required), versioning, and audit. Rolling back to a prior version is recorded as a forward change, not a deletion of history.

6.5 Multi-Tenancy in the Frontend The frontend does not implement multi-tenancy. It passes the authenticated user’s identity with every API call. The API returns only what the caller can access.

Search results are pre-filtered. A search for tasks returns only tasks the caller can see — filtered by role permissions (layer 1) and group membership (layer 2). The frontend does not need to add tenant filters to queries. The API handles it.

Field-level omissions. When access classification hides fields, the API response includes the field name with a metadata flag: `{"field": "salary", "access": "restricted", "value": null}`. The frontend renders this appropriately — “Access restricted” rather than an empty field, so the user knows data exists but is not visible to them.

Group-scoped navigation. The frontend can query the user’s group memberships and render navigation accordingly. A user who is a member of three project groups sees three projects. Adding the user to a fourth group (a change set on the group membership entity) immediately makes the fourth project visible — no frontend deployment, no configuration change, no cache invalidation.

7. Application Patterns by Type

Each architecture position has a distinct construction pattern. The patterns below describe what to build, what OpsDB provides, and where the development effort concentrates.

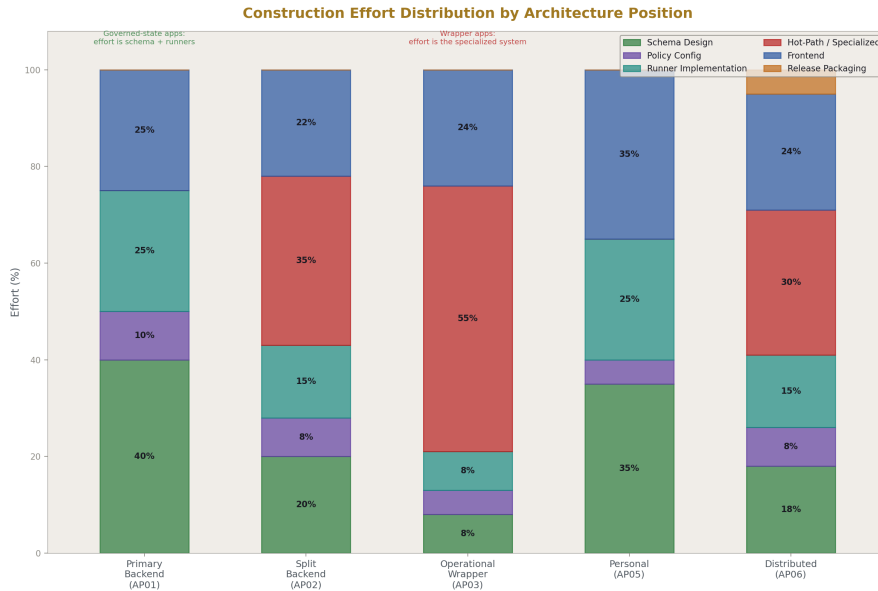


Figure 5: Fig. 4: Effort Distribution — schema-dominant for primary backend, specialized-system-dominant for wrapper, showing the crossover.

7.1 Governed-State-Dominant Pattern **Architecture position:** OpsDB is the primary backend. No hot path. 95%+ of the application is governed state management.

What you build: Schema YAML files for all domain entities. Policy rows for approval rules, access control, retention, and lifecycle rules. 3–6 runners for notification, integration, reporting, and domain computation. A frontend that consumes the API.

What OpsDB provides: The entire API layer — CRUD, search, change management, versioning, audit. Authentication and five-layer authorization. Schema validation and bound validation. Optimistic concurrency. Point-in-time state reconstruction. Retention enforcement.

Where effort concentrates: Domain analysis and schema design (60% of backend effort). Runner implementation (30%). Policy configuration (10%). The frontend is a separate effort of conventional size — OpsDB does not reduce frontend work, only backend work.

Representative applications: Project management, CRM, HR platform, compliance platform, inventory management, procurement, legal case management, education platform, personal data platform.

Construction sequence:

1. Domain analysis: enumerate entities, relationships, lifecycles, policies
2. Schema design: write YAML files for all entities

3. Loader run: generate database structure
4. Seed data: create initial policy rows, approval rules, role definitions, group memberships
5. API verification: create test entities, verify constraints, test approval flows
6. Runner implementation: build domain runners (notification, integration, computation)
7. Frontend development: build UI consuming the API
8. Policy tuning: adjust approval rules, access control, and retention as real usage reveals needs

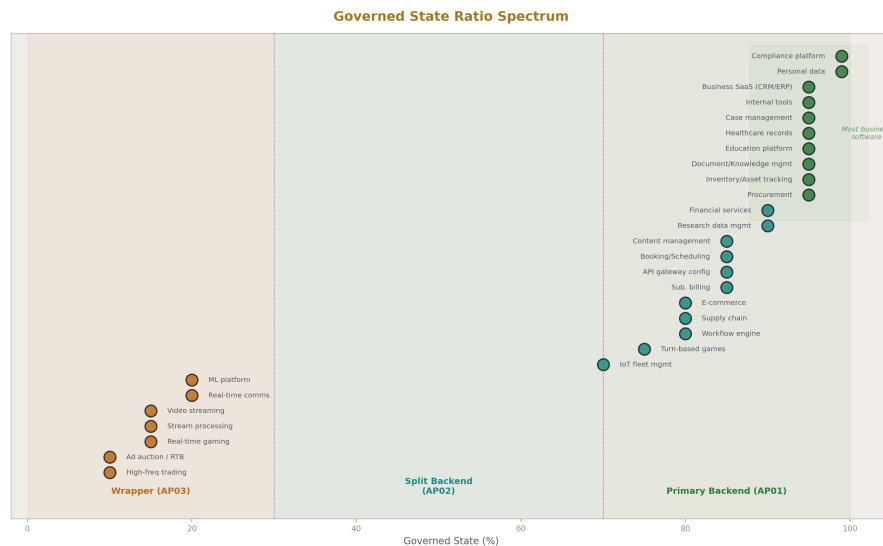


Figure 6: Fig. 7: Governed State Spectrum — most business software clusters at 95%+ governed state, with clear region boundaries at architecture positions.

7.2 Split-Backend Pattern **Architecture position:** OpsDB governs most state. A specialized system handles a bounded hot path. Runners bridge the two. 70–90% governed state.

What you build: Schema YAML for the governed portion (which is most of the data model). The hot-path service as a conventional system optimized for its specific requirement. Configuration push runners that format governed state for the hot-path service. Observation pull runners that bring results back. A frontend that consumes both the OpsDB API (for governed operations) and the hot-path service API (for real-time operations).

What OpsDB provides: Everything from the governed-state pattern, plus the runner infrastructure for bridging. The hot-path service gets its configuration from OpsDB via

runners and operates independently.

Where effort concentrates: The hot-path service (40–50% of total backend effort). Schema design for the governed portion (20–25%). Runner bridges (15–20%). Policy configuration (5–10%).

Representative applications: E-commerce (cart/checkout hot path), booking systems (reservation contention hot path), subscription billing (payment processing hot path), IoT fleet management (telemetry ingestion hot path), content management (content delivery hot path).

Construction sequence:

1. Domain analysis: identify governed entities AND hot-path requirements
2. Schema design for governed portion
3. Hot-path service design and implementation
4. Runner bridge design: what configuration flows out, what results flow back
5. Configuration push runner implementation
6. Observation pull runner implementation
7. Loader run and API verification for governed portion
8. Integration testing: verify the bridge — config changes in OpsDB propagate to hot-path service, results flow back as observations
9. Frontend development: consuming both APIs

The bridge pattern in detail:

The hot-path service reads configuration from a local cache that runners populate. The cache refresh cycle is configured in the runner spec — every 60 seconds, every 5 minutes, on-demand via a trigger. The hot-path service never calls the OpsDB API directly at run-time. If OpsDB is unavailable, the hot-path service continues with cached configuration. When OpsDB returns, the runner catches up.

OpsDB governed entities

- config push runner (reads entities, transforms, pushes)
- hot-path service local cache
- hot-path service processing

hot-path service results

- observation pull runner (reads results, transforms, writes)

→ OpsDB observation cache

→ search API (for dashboards, reports, reconciliation)

The observation pull runner writes results as observation cache entries with freshness timestamps. A reconciler runner can compare the observation cache against governed entities to detect discrepancies — “OpsDB says the customer’s plan is Premium but the billing engine charged them at the Basic rate.”

7.3 Operational Wrapper Pattern **Architecture position:** OpsDB wraps a specialized system with governance. The specialized system is the application. OpsDB manages configuration, accounts, policies, compliance, and audit. 10–30% governed state.

What you build: Schema YAML for the governance wrapper — accounts, configuration, policies, audit metadata. The specialized system as a purpose-built application (this is most of the engineering effort). Configuration push runners. Observation pull runners. A frontend that primarily interacts with the specialized system, with OpsDB-backed administration and compliance views.

What OpsDB provides: Governed management of configuration and policies. Audit trail for configuration changes. Compliance evidence production. Account management with full authorization model. The specialized system operates independently.

Where effort concentrates: The specialized system (60–70% of total effort). Schema design for the wrapper (10%). Runner bridges (10%). Frontend for the specialized system (10–15%). Admin/compliance frontend consuming OpsDB (5%).

Representative applications: Real-time communication (message bus is the system), stream processing (event processor is the system), ML platforms (training/inference is the system), video streaming (transcoding/delivery is the system), high-frequency trading (matching engine is the system).

Construction sequence:

1. Specialized system design and implementation (the primary engineering work)
2. Identify what configuration and operational data should be governed
3. Schema design for the governance wrapper
4. Runner bridge implementation
5. Admin frontend consuming OpsDB API
6. Compliance and audit views consuming OpsDB audit log and evidence records

7.4 Personal AppDB Pattern **Architecture position:** OpsDB is the primary backend on personal hardware. Single user. Auto-approve everything. Minimal runner population.

What you build: Schema YAML for personal data domains. Personal runners for external integrations (weather, fitness, banking, RSS). A simple frontend — a mobile app, a web interface, a CLI.

What OpsDB provides: Everything from the governed-state pattern, with governance features that simplify to invisibility. Change management auto-approves. Authorization is a single role. Audit logging runs silently in the background. Versioning and search are the primary value — you never lose data, and you can find anything.

Where effort concentrates: Schema design for personal domains (40%). Personal runners for integrations (30%). Frontend (30%).

Graduated formalization pattern:

Start with a generic collection entity that has a discriminated JSON payload. Track recipes, books, plants, and workout sessions as collection items with different payload schemas. When a domain becomes important enough to warrant proper schema treatment — your recipe collection grows from casual tracking to a serious reference — write a proper schema for that domain. A migration runner reads the generic collection items of that type and writes them as proper entities through change sets. The generic collection stays for casual tracking. The proper schema provides structured fields, relationships, typed queries, and specific runners.

```
# Generic collection for casual tracking

name: collection

domain: personal

fields:
  - name: name

type: varchar

length: 255

nullable: false

  - name: collection_type

type: enum

values: [recipe, book, plant, workout, wine, movie, travel]

nullable: false

  - name: item_data_json

type: json
```

```
nullable: false

- name: tags

type: varchar

length: 1024

nullable: true
```

As the personal AppDB matures, some collections graduate to proper schemas. Others stay generic indefinitely. The decision to graduate is driven by pain — when the generic schema’s lack of typed fields, explicit relationships, or specific runners becomes a friction, it is time to formalize. This is the correct use of iterative discovery within the OpsDB architecture — the generic collection is the sandbox, and graduation to a proper schema is the point where the domain is understood well enough for comprehensive design.

7.5 Distributed Application Pattern **Architecture position:** The application is packaged for deployment by others. The development team’s AppDB is the prototype. Each deployment is a separate DOS.

What you build: The prototype AppDB — schema, policies, runners, documentation. A release packaging process that produces a deployable artifact containing the schema YAML, runner binaries, default policy configurations, and operational documentation. A schema upgrade path from version N to version N+1 using the additive evolution rules.

What OpsDB provides: The deployment infrastructure. Each deploying organization runs their own OpsDB instance, loads the release schema, configures their own policies, and runs the packaged runners. The gate pipeline, versioning, audit, and authorization work identically in every deployment.

Release versioning:

Each release includes a schema version identifier. Schema changes between releases follow the additive evolution rules — new fields, new entities, widened constraints, no deletions, no renames, no type changes. Upgrading from version N to N+1 means applying the schema diff through the schema change set mechanism. The loader computes the diff between the deployed schema and the release schema, generates the additive changes, and applies them through a schema change set.

Release 1.0:

```
entities: project, task, comment, label

fields: (initial set)

policies: (default approval rules, retention policies)
```

Release 1.1:

entities: project, task, comment, label, milestone (added)

fields: task.estimated_hours (added), task.priority max widened from 5 to 10

policies: (default + milestone approval rules)

The upgrade from 1.0 to 1.1 adds the milestone entity, adds the estimated_hours field to task (nullable, so existing rows are unaffected), and widens the priority range. No existing data is affected. No existing consumers break. The deploying organization reviews the schema change set, approves it, and the schema executor applies it.

Per-deployment policy configuration:

The release includes default policies — default approval rules, default retention policies, default access control configuration. The deploying organization modifies these through standard change sets to match their requirements. A healthcare organization deploying a patient management AppDB sets retention policies to match medical record retention law. A small team deploying a project management AppDB sets all approval rules to auto-approve. The schema is shared. The policies are local.

8. AI-Assisted Construction Methods

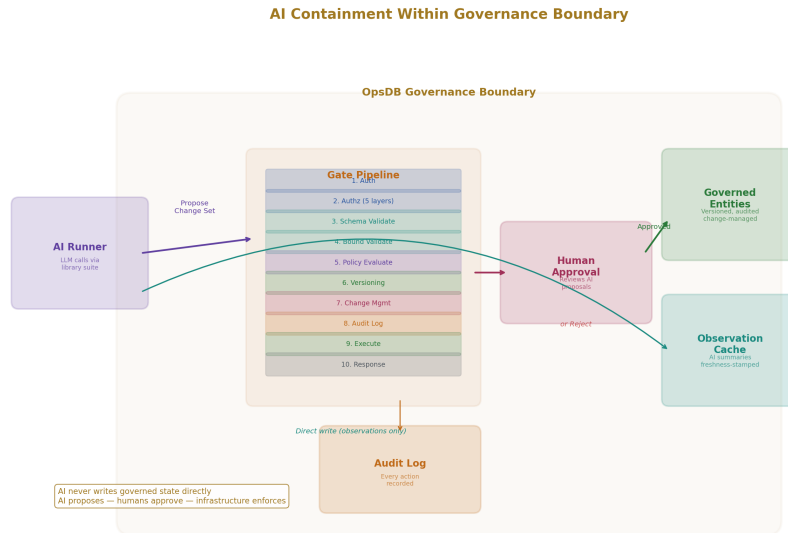


Figure 7: Fig. 6: AI Containment — AI proposes change sets through the gate pipeline, writes observations directly, humans approve governed changes.

The AppDB architecture is well-suited for AI assistance at every stage of construction. The closed vocabulary, mechanical verification, small artifact size, and governed change management create an environment where AI is productive and contained.

8.1 AI as Schema Generator An AI can generate schema YAML from a domain description. The prompt pattern:

1. Describe the domain in plain language — what entities exist, what their relationships are, what fields they have
2. Specify the schema conventions — singular names, lowercase underscores, FK naming, type suffixes
3. Ask for YAML output conforming to the loader's expected format

The AI generates YAML files. The loader verifies them mechanically — structural correctness, naming conventions, FK target existence, type validity, constraint completeness. If the loader rejects the output, the error message identifies the specific violation. The AI can be prompted to fix the violation, or the developer can fix it manually.

This works well because the output format is constrained and verifiable. The AI cannot hallucinate a new field type — only nine exist. It cannot hallucinate a new constraint mechanism — only six exist. It cannot generate structurally invalid YAML that passes

the loader — the loader catches every structural violation. The worst case is a schema that is structurally valid but domain-incorrect — wrong field types, wrong relationships, missing entities. These errors are visible in the YAML and correctable through human review.

8.2 AI as Runner Generator An AI can generate runner code from a runner spec description. The prompt pattern:

1. Describe the runner’s purpose — what it reads, what it computes, what it writes
2. Specify the get-act-set structure
3. Specify the library suite calls available
4. Ask for Go code using the library suite

The AI generates 150–300 lines of runner code. The code is small enough for complete human review. The library suite handles all infrastructure concerns, so the AI only generates domain logic — the act phase computation. Errors in the generated code are contained by the runner’s scope declaration — even incorrect runner code cannot affect data outside its declared scope because the API gate and library suite enforce boundaries.

8.3 AI as Change Proposer An AI operating as a system participant — an AI runner — reads governed state, makes domain decisions, and proposes change sets through the standard API. The AI is authenticated as a service account with declared capabilities and targets. Its proposals go through the same gate pipeline as human proposals.

Policy configuration for AI proposers:

Create approval rules that match on the AI service account’s identity. Low-risk AI proposals (updating observation cache summaries, categorizing items, computing derived values) can auto-approve. Medium-risk AI proposals (updating governed entity fields based on reconciliation) require human review. High-risk AI proposals (modifying policies, access control, financial data) require multiple human approvers.

The approval rules are data rows. Adjusting how much autonomy the AI gets means changing policy rows through governed change sets. The decision of “how much should we trust the AI” is itself a governed, versioned, auditable decision.

Containment:

The AI service account’s runner scope declarations limit what it can read and write. The gate pipeline validates every AI-proposed change set against the same constraints as human proposals — schema validation, bound validation, policy evaluation, authorization. If the AI proposes invalid changes, they are rejected at the gate. If the AI proposes changes outside its scope, they are rejected at authorization. If the AI proposes changes that require approval, they queue for human review.

The audit trail records every AI proposal — accepted, rejected, pending. The version history preserves the state before and after every AI-applied change. Rollback is a governed

change set that restores prior values. The AI cannot make irreversible changes because the system has no irreversible changes — every state is reconstructable from version history.

8.4 AI Observation Generation AI generates derived observations over governed data, as described in Section 5.8. The key points for construction:

When to use AI observations:

Incident summarization — the AI traverses the dependency chain, recent changes, alert fires, and produces a human-readable summary of what happened and what is affected.

Data classification suggestions — the AI reads entity content and proposes access classification levels. The proposals are observations in the cache. A human reviews the suggestions and applies them through governed change sets.

Anomaly narrative — the AI reads observation cache metrics and produces natural-language descriptions of anomalies: “CPU utilization on service X increased 40% over the past hour, coinciding with a deployment change set applied at 14:30.”

Compliance gap analysis — the AI reads the compliance regime entities, the evidence records, and the governed entity configurations, and produces a narrative identifying gaps: “No evidence record exists for backup verification on database Y for the past 30 days, which is required by SOC2-Availability.”

When not to use AI observations:

For factual queries that the search API answers directly. “How many tasks are overdue?” is a search query, not an AI observation. For computations that must be exact — financial totals, inventory counts, compliance metrics. These are custom domain runners with deterministic logic, not LLM calls with probabilistic output.

9. Construction Method Catalog

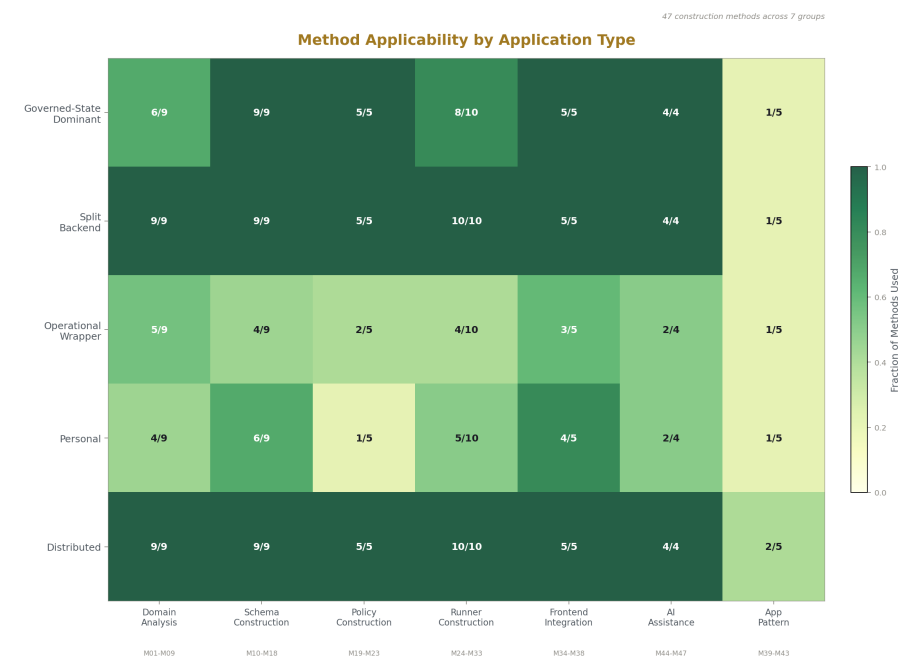


Figure 8: Fig. 2: Method Applicability — 47 methods across 7 groups, showing universal core vs position-specific methods.

Each method in this paper is cataloged below for reference. The catalog provides the method’s applicability, inputs, and outputs.

```
methods(id|name|section|applies_to|inputs|outputs)

# Domain Analysis Methods

M01|Entity Enumeration|2.1|all positions|domain knowledge|entity list with i
M02|Field Enumeration|2.2|all positions|entity list|field list with types, co
M03|Relationship Classification|2.3|all positions|entity list|classified rel
M04|Lifecycle Identification|2.4|entities with state|entity list|state enums
M05|Policy Enumeration|2.5|all positions|entity list + relationships|approva
```

M06|Schedule Enumeration|2.6|all positions|domain knowledge|recurring operations

M07|External Integration Enumeration|2.7|split + wrapper|domain knowledge|authentication

M08|Hot Path Identification|2.8|split + wrapper|domain knowledge + performance

M09|Architecture Position Selection|2.8|all|domain analysis results|position

Schema Construction Methods

M10|Entity Design|3.1|all positions|entity from M01 + fields from M02|entity

M11|Type Selection|3.2|all positions|field from M02|type + constraints from c

M12|Constraint Selection|3.3|all positions|field from M02|bounds, lengths, e

M13|Discriminator Pattern|3.4|entities with typed variants|entity with subtyp

M14|Bridge Table Design|3.5|many-to-many + polymorphic relationships|relatio

M15|Hierarchy Design|3.6|self-referential entities|entity with parent-child relationship|self-FK + depth field + traversal runner

M16|Governance Field Configuration|3.7|entities needing access control|access

M17|Draft Mode Configuration|3.8|interactive editing entities|property analy

M18|Schema Organization|3.9|all positions|complete entity set|directory stru

Policy Construction Methods

M19|Approval Rule Design|4.1|all governed entities|entity types + sensitivity

M20|Access Control Design|4.2|all positions|data sensitivity map|roles, group

M21|State Transition Rules|4.3|entities with lifecycles|transition graph from

M22|Retention Policy Design|4.4|all governed entities|regulatory + operation

M23|Cross-Field Invariant Rules|4.5|entities with conditional constraints|do

Runner Construction Methods

M24|Runner Design|5.1|all runners|domain logic requirement|runner spec with i

M25|Puller Runner|5.2|external data import|authority + external API|observat

M26|Reconciler Runner|5.3|desired vs observed correction|governed entities +

M27|Verifier Runner|5.4|scheduled verification|schedules + targets|evidence

M28|Notification Runner|5.5|state transition alerting|change sets + schedule

M29|Configuration Push Runner|5.6|hot path config delivery|governed entities

M30|Observation Pull Runner|5.7|hot path result collection|external system re

M31|AI Observation Runner|5.8|LLM-derived observations|governed entities + c

M32|Custom Domain Runner|5.9|domain-specific computation|governed entities +

M33|Server Runner Configuration|5.10|runner thread management|runner specs|t

Frontend Integration Methods

M34|Authentication Integration|6.1|all frontends|IdP configuration|SSO token

M35|Read Patterns|6.2|all frontends|search API|list views, detail views, acti
in-time views

M36|Write Patterns|6.3|all frontends|change set API|standard writes, auto-
approved, approval-required, creation, bulk, emergency

M37|Draft Mode Integration|6.4|draft-mode entities|draft mode API|autosave +

M38|Multi-Tenancy Integration|6.5|multi-user applications|authorization API
filtered results, field omissions, group-scoped navigation

Application Pattern Methods

M39|Governed-State-Dominant Construction|7.1|primary backend position|full o

M40|Split-Backend Construction|7.2|split backend position|domain analysis +

M41|Operational Wrapper Construction|7.3|wrapper position|specialized system

M42|Personal AppDB Construction|7.4|personal scale|personal data domains|per

M43|Distributed Application Construction|7.5|packaged applications|prototyp
deployment policy

AI-Assisted Methods

M44	AI Schema Generation 8.1 all positions domain description + conventions	verifiable YAML files
M45	AI Runner Generation 8.2 all runners runner spec description + library su	300 line runner code
M46	AI Change Proposer Configuration 8.3 governed entities AI runner spec + a	
M47	AI Observation Configuration 8.4 governed entities with summary needs ru	

10. Decision Trees

For each major construction decision, a decision tree identifies the method to apply.

10.1 Architecture Position Selection

Does the application have processing that cannot tolerate the gate pipeline la

|— No → Is it a personal or single-user application?

| |— Yes → Personal AppDB (M42)

| |— No → Is it intended for distribution as a deployable package?

| |— Yes → Distributed Application (M43) with Governed-State-Dominant (M39)

| |— No → Governed-State-Dominant (M39)

|— Yes → What percentage of the data model is governed state?

|— 70%+ → Split-Backend (M40)

|— 10-70% → Operational Wrapper (M41)

|— <10% → OpsDB may not be the right tool; evaluate metadata-only position

10.2 Entity Design Selection

Does the entity have subtypes with different field sets?

- |— Yes → Discriminator Pattern (M13)
- └─ No → Does the entity relate to multiple different entity types polymorphically?
 - |— Yes → create bridge tables per target type (M14)
 - └─ No → Does the entity have a parent-child hierarchy?
 - |— Yes → Self-referential FK + depth field (M15)
 - └─ No → Standard Entity Design (M10)

10.3 Runner Kind Selection

What does the runner do?

- |— Imports data from external system → Puller (M25)
- |— Compares desired vs observed and corrects → Reconciler (M26)
- |— Verifies conditions and produces evidence → Verifier (M27)
- |— Dispatches notifications on state transitions → Notification (M28)
- |— Pushes governed config to external system → Config Push (M29)
- |— Pulls results from external system → Observation Pull (M30)
- |— Generates LLM-derived summaries → AI Observation (M31)
- └─ Domain-specific computation → Custom Domain (M32)

10.4 Gating Mode Selection

What does the runner write?

- |— Observation cache (external state, metrics, AI summaries)
 - | → Direct write (authenticated, validated, audited, no change set)
- |— Governed entity changes with low risk
 - | → Auto-approved change set (change set created, auto-transitions)

- |— Governed entity changes with high risk or compliance requirements
- | → Approval-required change set (routes to human approvers)
- |— Mixed (different targets with different stakes)
- Per-target gating (direct write for observations, approval for governed)

10.5 Governance Flag Decision

Is this entity used for interactive editing where per-keystroke versioning creates noise?

- |— No → Full governance (default)
- |— Yes → Does the entity hold compliance-relevant data?
 - |— Yes → Full governance (compliance requires per-write auditability)
 - |— No → Does the entity hold financial data?
 - |— Yes → Full governance (financial data requires per-write versioning)
 - |— No → Property analysis: which properties are weakened by draft mode?
 - |— Per-write versioning → acceptable if committed versions suffice
 - |— Per-write audit → acceptable if committed version audit suffices
 - |— Change management → acceptable if auto-commit governance suffices
- Enable draft mode flags + document tradeoff in schema notes (M17)

10.6 AI Assistance Selection

What task are you performing?

- |— Schema design from domain description → AI Schema Generation (M44)
- |— Runner implementation from spec → AI Runner Generation (M45)

- |— Automated domain decisions that modify governed state → AI Change Proposer
 - |— Derived summaries over governed data → AI Observation (M47)
 - |— Domain analysis, policy design, architecture decisions → Human judgment
- (AI can assist with enumeration but the decisions are engineering)
-

11. Cross-Reference to Taxonomy

Each construction method connects to the underlying taxonomy's mechanisms, properties, and principles.

`method_to_taxonomy(method|mechanisms|properties|principles)`

M01|AF1 (Storage)+AF13 (Representation) |P04 (Consistency) |R05 (Comprehensive)+

M02|AF13 (Representation) |P04 (Consistency)+P08 (Determinism) |R01 (Data Primacy

M03|AF1 (Storage)+AF4 (Selection) |P04 (Consistency)+P05 (Integrity) |R05 (Compre

M04|NM01 (State Machine Evaluator) |NP01 (Lifecycle Integrity) |NR02 (Rules as Da

M05|AF2 (Gating)+AF9 (Coordination) |P06 (Authenticity)+P07 (Confidentiality)+P

M06|AF8 (Lifecycle)+NM04 (Temporal Projection) |NP02 (Temporal Consistency) |R01

M07|AF5 (Information Movement) |P11 (Availability)+P19 (Failure Transparency) |R

M08|AF5+AF10+AF11+AF12 |P10 (Liveness)+P11 (Availability)+P12 (Boundedness) |R1

M10|AF1+AF13 |P04+P08 |R01+R03 (Convention over Lookup)

M13|AF13+NM02 (Rule Engine) |P04+P08 |R01+R04 (0/1/Infinity)

M14|AF1+AF4 |P04+P05 |R05+R06 (One Way)

M17|varies per flag|weakens P20 (Observability)+P21 selectively|NR03 (Separate

M19|AF2+AF9 |P06+P21+NP05 (Non-repudiation) |R18+NR02

M20|AF2 |P06+P07+P13 (Isolation) |R09 (Fail Closed)+R18

M21 | NM01 | NP01 | NR02

M24 | AF6 (Control Loop) + AF7 (Sensing) | P01 (Idempotency) + P10 + P12 | R07 (Idempotent Triggered) + R11 (Bound Everything)

M25 | AF5 + AF7 | P03 (Durability) + P11 | R08 + NR01 (Separate Domain from Infra)

M26 | AF6 + AF7 | P01 + P04 + P09 (Convergence) | R08 + R07

M27 | AF7 + AF8 | P20 + P21 | R21 (Make State Observable)

M28 | AF5 | P10 + P11 | R18 + R19 (Push Decision Down)

M29 | AF5 + AF10 | P11 + P19 | R17 + R20 (Push Work Down)

M30 | AF5 + AF7 | P03 + P11 | R17 + R08

M31 | AF5 + AF10 + NM03 (Accumulator) | P11 + SP04 (Domain Opacity) | NR01 + SM04 (Semantic)

M32 | AF10 + NM02 + NM03 | NP03 (Computational Correctness) | NR01 + NR02

M33 | AF12 (Allocation) + AF9 | P10 + P12 + P13 | R04 + R11 + SM06 (Scene as Isolation)

M39 | all governed mechanisms | all governed properties | all universal principles

M40 | governed + AF5 + AF11 | governed + P11 + P19 | R14 + R17 + R19

M41 | AF1 + AF2 + AF5 + AF7 | P06 + P07 + P20 + P21 | R14 + R17 + NR01

M44 | AF13 + SM04 | SP01 (Vocabulary Closure) + SP02 (Structural Security) | NP02 (Vocab)

M46 | AF2 + AF6 + SM07 (Structured Trace) | P01 + P21 + SP03 (Hot-Swap Safety) | NP01 (Security by Anatomy) + NP04 (Infra Fix Protects All)

12. Boundaries and Limitations

The construction methods in this paper apply within the OpsDB abstraction boundary. Several limitations constrain what can be built.

The closed vocabulary is a real constraint. Nine types, six constraints, three modifiers. If your domain requires a constraint that the vocabulary cannot express — a field that must match a specific pattern, a constraint that depends on external state, a type not in the nine — you cannot express it in the schema. Cross-field invariants go to policy rows. Complex validation goes to runner logic that runs before proposing change sets. But some validation requirements may not fit cleanly.

Schema evolution is additive only. You cannot delete fields, rename entities, or change types. The six-step duplication pattern handles type changes but at the cost of permanent tombstone fields. Over years, schemas accumulate deprecated fields. This is a real cost — the schema grows monotonically. The tradeoff is that consumers, version history, audit logs, and runner logic never break from schema changes.

The gate pipeline adds latency. Every write passes through ten steps. For human-pace operations this is invisible. For high-frequency automated writes — thousands per second — the pipeline is the bottleneck. Observation cache tables with appropriate governance flags reduce this for observation-like data. For truly high-frequency requirements, a hot-path service outside OpsDB is necessary.

OpsDB is not a compute platform. Runners perform computation, but heavy computation — ML training, video transcoding, scientific simulation — belongs in specialized systems. OpsDB governs the configuration and collects the results. The computation itself runs elsewhere.

Discovery-phase prototyping is costly. The additive-only evolution rules penalize schema experimentation. If you do not yet know your data model, the cost of guessing wrong is permanent tombstone fields. The graduated formalization pattern (generic collection → proper schema) mitigates this for some domains. For others, prototyping in a less rigid system and migrating to OpsDB once the domain stabilizes is the right approach.

The frontend is your responsibility. OpsDB provides the data substrate. It does not render pages, serve static assets, manage client-side state, handle responsive layout, or implement user interaction patterns. The frontend is a conventional development effort of conventional size. OpsDB reduces backend work by an order of magnitude. It does not reduce frontend work.

13. Summary

This paper enumerates 47 construction methods for building applications with OpsDB Application Architecture. The methods cover domain analysis (9 methods), schema construction (9 methods), policy construction (5 methods), runner construction (10 methods), frontend integration (5 methods), application patterns by type (5 methods), and AI-assisted construction (4 methods).

The methods compose. A governed-state-dominant application uses methods M01–M06 for domain analysis, M10–M18 for schema construction, M19–M23 for policy construction, M24–M28 plus M31–M32 for runners, M34–M38 for frontend integration, and M44–M47 for AI assistance. A split-backend application adds M08, M29, M30, and M40. A personal AppDB uses M42 with graduated formalization. A distributed application adds M43 with release versioning.

The construction sequence for every application type follows the same fundamental pattern: analyze the domain comprehensively, declare it as schema, verify it through the

loader and API, implement domain-specific logic as runners, and connect a frontend. The infrastructure — validation, authorization, versioning, change management, audit, search, retention — exists from the first entity definition. The developer’s effort concentrates on understanding the domain and declaring it correctly. The infrastructure handles everything else.

The methods are grounded in the three-axis taxonomy from the infrastructure specification: mechanisms determine what the application does with data, properties determine what guarantees it needs, and principles determine how it is assembled. Each method in the catalog traces to specific mechanisms, properties, and principles. The construction decisions are engineering decisions made on explicit axes, not fashion-driven choices copied from blog posts.

The architecture rests on data primacy: the schema is data, the policies are data, the business rules are data, the runner configurations are data, the AI observations are data. Code is fixed infrastructure that interprets data, or small runners that make domain decisions within declared bounds. The developer’s long-lived artifact is the schema. Everything else — runners, frontends, infrastructure versions — is replaceable. The data survives.

Appendix A: Entity Design Pattern Reference

ID	Pattern	Description	When to Use	YAML Elements	Example
EP01	Standard Entity	Single entity type with typed fields and constraints	Most entities; one identity, one set of attributes	name, domain, versioned, fields with types and constraints	task, project, customer, invoice
EP02	Discriminator Entity	Entity with subtype-specific typed JSON payload	Entity has variants with different field sets sharing common structure	Discriminator enum field + * data json field + per-type JSON schemas	monitor (prometheus query, http probe), schedule (cron, rate, calendar)
EP03	Bridge Entity	Join table connecting two entity types	Many-to-many relationships; polymorphic attachment	Two mandatory FK fields + optional relationship metadata fields	task label, project member, service ownership

ID	Pattern	Description	When to Use	YAML Elements	Example
EP04	Polymorphic Bridge Set	Multiple bridge tables connecting one source to multiple targets	One entity type attaches to several different entity types	One bridge table per source-target pair, each with two mandatory FKs	task comment + project comment + milestone comment
EP05	Hierarchical Entity	Self-referential entity with parent chain	Tree structures: org units, categories, locations, task/subtask	Nullable self-FK (parent * id) + depth int field + hierarchy runner	category, organizational unit, location
EP06	Governance Scoped Entity	Entity with per-row access control	Entities visible only to specific groups or with field-level sensitivity	requires group FK + access classification FK	patient record, salary data, classified document
EP07	Draft Mode Entity	Entity with relaxed interim governance	Interactive editing where per-keystroke versioning creates noise	draft mode flags (autoversion disabled, edit latest version, audit logs disabled)	document, note, configuration draft
EP08	Observation Cache Entity	Non-governed entity holding external or derived state	Imported external data, metrics, AI summaries, computed caches	versioned: false; written by scoped runners; freshness timestamp field	observation cache state, observation cache metric

ID	Pattern	Description	When to Use	YAML Elements	Example
EP09	Evidence Record Entity	Append-oriented entity recording verification results	Scheduled checks producing pass/fail with supporting data	Discriminator for evidence type + result enum + evidence data json	evidence record (backup verification, certificate validity)
EP10	Schedule Entity	Entity representing recurring or deadline-driven work	Anything that recurs, expires, or has temporal triggers	Discriminator for schedule type + schedule data json + last/next execution fields	schedule (cron expression, rate based, calendar anchored)
EP11	Policy Entity	Entity expressing behavioral rules evaluated by gate pipeline	Approval rules, transition rules, invariants, retention, access control	Discriminator for policy type + policy data json	policy (approval rule, state transition, semantic invariant)
EP12	Authority Entity	Entity representing an external system connection	External systems the application integrates with	Discriminator for authority type + authority data json + connection params	authority (payment processor, identity provider, cloud api)
EP13	Runner Spec Entity	Entity declaring a runner's configuration and bounds	Every runner in the system	Discriminator for runner type + runner data json with bounds, scope, trigger	runner spec (puller, reconciler, verifier, ai observation)

ID	Pattern	Description	When to Use	YAML Elements	Example
EP14	Generic Collection Entity	Catch-all entity for casual tracking with graduated formalization	Personal AppDB; early-stage domain exploration	collection type enum + item data json + tags	collection (recipe, book, plant, workout)

Appendix B: Field Type Decision Matrix

Condition	Type	Constraints to Declare	Notes
Whole number with domain bounds	int	min, max	Always declare both bounds; use for currency in minor units
Decimal number with domain bounds	float	min, max	Always declare both bounds; avoid for exact financial arithmetic
Short bounded string (names, titles, codes)	varchar	length	63 for identifiers, 255 for names, 1024 for descriptions/URLs
Long unbounded text (prose, markdown, notes)	text	none	Use sparingly; most strings have natural bounds
True/false present state	boolean	none	Prefix is (is active, is published)
True/false past event	boolean	none	Prefix was (was approved, was escalated)
Point in time	datetime	none	Suffix time; stored UTC; timezone handling in runners
Calendar day without time	date	none	Suffix date; use when time component is meaningless

Condition	Type	Constraints to Declare	Notes
Constrained value from fixed set	enum	values[]	Values can be added never removed; plan initial set carefully
Subtype-specific structured data	json	none (validated by discriminator-selected JSON schema)	Always pair with discriminator enum field
Reference to another entity	foreign key	references (target table)	Name as referenced table id or role referenced table id

Appendix C: Constraint Selection Matrix

Constraint Type	Applies To	Declaration	Enforcement Point	Evolution Rule
Numeric range	int, float	min, max	Gate step 4 (bound validation)	Can widen (lower min or raise max); never narrow
String length	varchar	length	Gate step 4	Can widen; never narrow
Enum value set	enum	values[]	Gate step 4	Can add values; never remove
Foreign key reference	foreign key	references	Gate step 4 (existence check)	Target table must exist; FK is permanent
Nullable	all types	nullable: true/false	Gate step 3 (schema validation)	Can change from not-null to nullable; never the reverse
Unique	all except text and json	unique: true	Gate step 4	Can add to empty table; cannot add with existing duplicates; cannot remove

Appendix D: Relationship Pattern Matrix

Pattern	Cardinality	FK		Naming	Example
		Implementation	Nullable		
Ownership	One-to-many mandatory	FK on child entity	false	parent table id	task.project id
Optional reference	One-to-many optional	FK on referencing entity	true	referenced table id	task.assignee user id
Multiple references to same target	One-to-many multiple	Multiple FKs with role prefixes	varies	role table id	contract.vendor company id, contract.client company id
Many-to-many	Many-to-many	Bridge table with two mandatory FKs	false (both)	table a id + table b id	task label (task id + label id)
Polymorphic attachment	One-to-many across types	One bridge table per target type	false (both)	source id + target id	task comment, project comment, milestone comment
Self-referential hierarchy	Tree	Nullable self-FK + depth field	true (parent)	parent table id + depth int	category.parent category id

Appendix E: Governance Field Reference

Field	Type	Purpose	Gate Layer	Default	When to Use
requires group	FK → ops group	Restrict entity visibility to group members	Layer 2 (per-entity governance)	null (no restriction)	Project-scoped data, team-scoped data, personal data

Field	Type	Purpose	Gate Layer	Default	When to Use
access classification	FK → data classification	Control field-level visibility by sensitivity	Layer 3 (per-field classification)	null (no restriction)	Salary data, medical details, financial details, PII
versioned: true	Schema flag	Enable change management and version history	Step 6 (versioning) + step 7 (change mgmt)	false	All entities where change history matters
autoversion disabled	Draft mode flag	Skip version row on interim saves	Step 6	false (version every write)	Interactive editing entities (documents, notes)
edit latest version	Draft mode flag	Write directly to current row, skip change set	Step 7	false (route through change set)	Interactive editing entities
audit logs disabled	Draft mode flag	Skip audit log on interim saves	Step 8	false (log every operation)	Interactive editing entities (use with caution)

Appendix F: Policy Type Reference

ID	Type	Purpose	Evaluated At	Match Criteria	Effect
PT01	approval rule	Determine who must approve a change set	Gate step 7	Entity type, field names, namespace, data classification, security zone, proposer role	Approval count + approver source + approver role
PT02	state transition	Declare valid state machine transitions	Gate step 5	Entity type, field, from value, to value	Accept or reject the transition
PT03	semantic invariant	Enforce cross-field conditional constraints	Gate step 5	Entity type, condition (field + operator + value), requirement (field + constraint)	Accept or reject with message
PT04	retention	Declare per-entity-type retention horizons	Reaper runner reads	Entity type	Version history days + observation cache days + audit log days
PT05	access control	Declare role-to-entity-type permissions	Gate step 2 layer 1	Role, entity type, operation (read/write)	Allow or deny
PT06	change management	Declare emergency review windows and SoD rules	Gate step 7	Entity type, change type	Review window hours + segregation of duties flag

ID	Type	Purpose	Evaluated At	Match Criteria	Effect
PT07	schedule governance	Declare scheduling constraints for runners and operations	Gate step 5	Runner type, schedule type	Allowed windows + blackout periods
PT08	security zone	Declare environment-level access restrictions	Gate step 2 layer 5	Zone, role, time of day, tenure	Allow or deny with additional conditions
PT09	data classification	Declare sensitivity levels and access grants	Gate step 2 layer 3	Classification level, role	Visible fields or hidden fields

Appendix G: Runner Kind Reference

ID	Kind	Purpose	Reads From	Writes To	Gating Mode	Trigger	Typical Size (lines)
RK01	Puller	Import external data	External API + runner spec + authority	observation cache *	Direct write	Scheduled (cron/rate)	200–400
RK02	Reconciler	Compare desired vs observed, correct drift	Governed entities + observation cache	change set or entity rows	Auto-approve or approval-required	Scheduled	200–350
RK03	Verifier	Check conditions, produce evidence	Schedule + target entities + prior evidence	evidence record	Direct write	Scheduled	150–250

ID	Kind	Purpose	Reads From	Writes To	Gating Mode	Trigger	Typical Size (lines)
RK04	Notification	Detect transitions, dispatch messages	change sets + schedules + on call + authority	observation cache (delivery receipts) + external channels	Direct write (observations)	Scheduled or event-driven	200–350
RK05	Change-set Executor	Apply approved change sets	Approved change sets	Entity rows + version rows	Post-approval direct	Event-driven (new approvals)	150–250
RK06	Reaper	Enforce retention policies	retention policy rows	Deletes/soft deletes on expired rows	Direct write	Scheduled	150–200
RK07	Config Push	Push governed state to external system	Governed entities + runner spec	External system config + observation cache (push receipts)	Direct write (observations)	Scheduled or on-change	200–350
RK08	Observation Pull	Pull results from external system	External API + runner spec + authority	observation cache *	Direct write	Scheduled	200–300
RK09	AI Observation	Generate LLM-derived summaries	Governed entities + context traversal	observation cache * (AI summaries)	Direct write	Scheduled or on-change	200–350

ID	Kind	Purpose	Reads From	Writes To	Gating Mode	Trigger	Typical Size (lines)
RK10	Reactor	Process external events	Webhook/runner + spec	Observation or change set	Varies (direct for obs, approval for entities)	Event-driven	200–300
RK11	Drift Detector	Find discrepancies without correcting	Desired entities + observation cache	change set (proposal only)	Approval-required	Scheduled	200–300
RK12	Scheduler	Manage execution timing for other runners	Schedule entities + runner specs	runner job entities	Auto-approve	Scheduled	150–250
RK13	Bootstrapper	Provision new entities from templates	Template config + trigger entity	Entity rows via change set	Auto-approve or approval	Event-driven	200–350
RK14	Custom Domain	Domain-specific computation	Governed entities + domain rules	Computed results as entities or observations	Varies	Varies	150–300

Appendix H: Gating Mode Decision Matrix

Mode	Change Set Created	Approval Routing	Audit Logged	Version Created	Use When
Direct write	No	No	Yes	No	Observation cache writes, evidence records, runner job updates, metrics, AI observations
Auto-approve	Yes	Auto-transitions to approved	Yes	Yes (if versioned)	Low-risk governed entity changes, routine automated updates, personal AppDB all changes
Approval-required	Yes	Routes to human approvers	Yes	Yes (on apply)	High-risk changes, financial data, access control, compliance-relevant, AI-proposed governed changes
Post-approval direct	No (already approved)	Already completed	Yes	Yes	Change-set executor applying previously approved change sets

Mode	Change Set Created	Approval Routing	Audit Logged	Version Created	Use When
Emergency	Yes (reduced)	Reduced approval count + mandatory flag	Yes (with emergency flag)	Yes	Break-glass situations with mandatory post-incident review

Appendix I: Schedule Type Reference

ID	Type	Payload Fields	Use When	Example
ST01	Cron expression	cron string, timezone	Fixed recurring schedule with specific timing	“0 2 * * *” — daily at 2 AM
ST02	Rate-based	interval seconds	Simple periodic execution	Every 300 seconds (5 minutes)
ST03	Event-triggered	trigger entity type, trigger field, trigger condition	Execution after specific state change	Run compliance scan 24h after any policy change
ST04	Calendar-anchored	anchor date, recurrence rule, timezone	Business calendar operations	Quarterly on first business day; annually on fiscal year start
ST05	Deadline-driven	deadline field, entity type, warning threshold days	Approaching deadline notification or verification	Certificate expires in 30 days; contract renewal due in 60 days
ST06	Manual	authorized roles, required evidence type	Human-initiated scheduled operations	Physical inspection; vendor review; tape rotation

Appendix J: Draft Mode Property Impact Matrix

Flag	Property Affected	Impact	Still Enforced	Risk Level
autoversion disabled	Per-write versioning	Interim states not individually recoverable	Committed version history intact	Low — recovery granularity reduced to committed versions
autoversion disabled	Point-in-time reconstruction	Reconstruction available at committed version granularity only	Committed version reconstruction O(1)	Low — acceptable for editing use cases
edit latest version	Change management (approval routing)	Interim saves bypass change set pipeline	Committed versions go through full change management	Medium — behavioral changes between commits are ungoverned
edit latest version	Attribution of interim changes	Individual interim saves not attributed to specific change sets	Committed versions fully attributed	Low — interim saves attributed via auth identity in request
audit logs disabled	Auditability	Interim saves not individually recorded in audit log	Committed versions produce audit entries	Medium — interim activity invisible to audit queries
audit logs disabled	Compliance evidence	Gap in audit trail between committed versions	Committed version audit entries satisfy per-version compliance	High for compliance-relevant data — do not use on regulated entities

Flag	Property Affected	Impact	Still Enforced	Risk Level
All three combined	Combined weakening	Interim editing is structurally validated and authorized but not versioned, change-managed, or audited	Auth + authz + schema validation + bound validation + policy evaluation	Acceptable only for non-regulated interactive editing entities

Appendix K: Schema Evolution Rule Reference

ID	Change Type	Allowed	Conditions	Mechanism	Risk
SE01	Add nullable field	Yes	Field must be nullable	Loader adds column with NULL	None — existing rows unaffected
SE02	Add entity type	Yes	No conditions beyond naming conventions	Loader creates new table	None — no existing consumers affected
SE03	Add enum value	Yes	New value appended to set	Loader updates check constraint	None — existing rows retain current values
SE04	Widen numeric range	Yes	New min $\leq$ old min AND new max $\geq$ old max	Loader updates check constraint	None — existing valid values remain valid
SE05	Widen string length	Yes	New length $\geq$ old length	Loader updates column length	None — existing valid values remain valid

ID	Change Type	Allowed	Conditions	Mechanism	Risk
SE06	Add index	Yes	No conditions	Loader creates index	Temporary — index build may lock table briefly
SE07	Delete field	Forbidden	Never	n/a	Breaks version history, audit log, consumers
SE08	Rename field or entity	Forbidden	Never	n/a	Breaks every consumer by name
SE09	Change field type	Forbidden	Never; use six-step duplication pattern	n/a	Breaks consumers expecting prior type
SE10	Narrow numeric range	Forbidden	Never	n/a	Existing rows may violate new bound
SE11	Remove enum value	Forbidden	Never	n/a	Existing rows may hold removed value
SE12	Tighten uniqueness	Forbidden	Never	n/a	Existing rows may violate new constraint
SE13	Add not-null field	Forbidden (unless empty table)	Only on empty table before data exists	Loader adds NOT NULL column	Breaks if table has existing rows without default

Duplication Pattern Steps

Step	Action	Duration	Notes
DP01	Add new field alongside old	Immediate	New field is nullable; old field unchanged
DP02	Begin writing to both fields	Deployment	All writers update both old and new fields
DP03	Migrate readers to new field	Gradual	Readers switch from old field to new field
DP04	Mark old field deprecated	Change set	Notes field documents deprecation
DP05	Continue writing both for safety period	Configurable	Ensures all readers have migrated
DP06	Stop writing to old field	Deployment	Old field becomes tombstone; never deleted

Appendix L: Library Suite Call Reference

ID	Library	Call	Purpose	Runner Phases	Handles
LC01	opsdb-api	search	Query governed entities with filters, joins, projection, pagination	get	Auth, pagination, rate limiting
LC02	opsdb-api	get entity	Fetch single entity with optional joins	get	Auth, field-level access filtering
LC03	opsdb-api	get entity at time	Reconstruct entity state at timestamp	get	Auth, version lookup
LC04	opsdb-api	get entity history	Fetch version chain for entity	get	Auth, pagination

ID	Library	Call	Purpose	Runner Phases	Handles
LC05	opsdb-api	submit change set	Propose field changes with reason	set	Auth, validation, scope checking, optimistic concurrency
LC06	opsdb-api	bulk submit	Propose multi-entity atomic changes	set	Auth, validation, scope checking, atomicity
LC07	opsdb-api	write observation	Write to observation cache tables	set	Auth, report key validation, scope checking
LC08	opsdb-api	emergency apply	Break-glass change with reduced approval	set	Auth, emergency role check, audit flagging
LC09	k8s	list resources	Query Kubernetes API for resources	get	Auth, pagination, rate limiting
LC10	k8s	apply manifest	Apply Kubernetes manifest	act	Auth, dry-run support, retry, conflict handling
LC11	k8s	watch events	Subscribe to Kubernetes event stream	get	Auth, re-connection, backpressure
LC12	cloud-aws	describe resources	Query AWS API for resource state	get	Auth, pagination, rate limiting, region handling

ID	Library	Call	Purpose	Runner Phases	Handles
LC13	cloud-aws	apply change	Modify AWS resource	act	Auth, retry, idempotency token, dry-run support
LC14	cloud-gcp	describe resources	Query GCP API for resource state	get	Auth, pagination, rate limiting, project scoping
LC15	cloud-gcp	apply change	Modify GCP resource	act	Auth, retry, operation polling
LC16	secrets	read secret	Fetch secret value from vault	act	Auth, lease management, caching
LC17	secrets	rotate secret	Generate and store new secret	act	Auth, atomic swap, old-version retention
LC18	notification	send email	Dispatch email via configured SMTP	act	Auth, retry, template rendering, rate limiting
LC19	notification	send webhook	Dispatch webhook to configured URL	act	Auth, retry, signature, timeout
LC20	notification	send chat	Dispatch message to chat platform	act	Auth, retry, rate limiting, thread management
LC21	logging	structured log	Emit structured log entry with correlation ID	all phases	Correlation propagation, level filtering, output formatting

ID	Library	Call	Purpose	Runner Phases	Handles
LC22	logging	metric	Emit metric data point	all phases	Metric naming, label propagation, buffering
LC23	git	clone repo	Clone git repository	get	Auth, shallow clone, sparse checkout
LC24	git	commit and push	Commit changes and push	act	Auth, retry, conflict detection
LC25	template	render	Render template with variable substitution	act	No logic execution; substitution only
LC26	llm	generate	Call LLM API with prompt and parameters	act	Auth, retry, timeout, token budget enforcement, circuit breaking
LC27	lifecycle	heartbeat	Report runner thread health	set	Cycle timing, error counting, observation
LC28	lifecycle	claim work	Claim work item via advisory lock	get	cache write Lock acquisition, conflict resolution, timeout
LC29	lifecycle	release work	Release work item claim	set	Lock release, completion recording

Appendix M: AI Runner Configuration Reference

Field	Type	Purpose	Example Values
model	varchar	LLM model identifier	“claude-sonnet”, “claude-haiku”
max tokens	int	Maximum response tokens from LLM	100, 500, 2000
character budget	int	Target character count for output	12, 80, 500, 2000, 5000
regeneration trigger	enum	When to regenerate observation	source entity change, scheduled, on demand
trigger schedule	varchar	Cron or rate for scheduled regeneration	“/15 * * *”, “rate:900”
target entity types	json array	Which entity types to summarize	[“incident”], [“task”, “project”]
context traversal	json array	Relationship paths to traverse for context assembly	[“incident -> affected service”, “incident -> change set”]
prompt template ref	varchar	Reference to prompt template entity	“incident summary v1”, “anomaly narrative v2”
observation cache table	varchar	Target observation cache table	“observation cache state”
observation key prefix	varchar	Prefix for observation cache keys	“ai summary incident”, “ai classify”
retry budget	int	Max LLM call retries per cycle	3
execution timeout seconds	int	Max time for LLM call including retries	60, 120
scope per cycle	int	Max entities to process per runner cycle	10, 50, 100

AI Observation Granularities

ID	Name	Character Budget	Use Case	Prompt Style	Audience
AG01	Ticker headline	80	Dashboard ticker, mobile notification header	Terse single-sentence	All users
AG02	Severity tag	12	Severity classification label	Single-word or short-phrase classification	Automated routing
AG03	Brief summary	500	Notification body, list view annotation	Concise paragraph with key facts	Operational staff
AG04	Standard summary	2000	War room display, incident detail view	Structured multi-paragraph with context	Incident responders
AG05	Detailed analysis	5000	Post-incident review, compliance narrative	Thorough analysis with dependency chain walk-through	Investigators and auditors
AG06	Structured classification	100	Data classification suggestion, category proposal	JSON-formatted classification with confidence	AI change proposer pipeline

Appendix N: Architecture Position Quick Reference

Position	Governed Ratio	Hot Path	OpsDB Role	Runner Bridge	Frontend Target	Example Applications
AP01 Primary backend	95–100%	None	Full backend	n/a	OpsDB API only	CRM, HR, compliance, project mgmt, personal data, education, legal, procurement, inventory
AP02 Split backend	70–90%	Bounded specific path	Governance + catalog	Config push + observation pull	OpsDB API + hot-path API	E-commerce, booking, billing, CMS, IoT fleet, API gateway config, workflow, turn-based games
AP03 Operational wrapper	10–30%	Dominant	Config + accounts + policies + audit	Config push + observation pull	Primarily hot-path; OpsDB for admin	Real-time chat, streaming, ML platform, video streaming, ad auction, HFT

Position	Governed Ratio	Hot Path	OpsDB Role	Runner Bridge	Frontend Target	Example Applications
AP04 Metadata manager	5–10%	Entire system	Structured pointers + operational metadata	Metadata sync	Specialized system; OpsDB for ops views	Time-series DB, search engine, graph DB, object store, message broker
AP05 Personal	99%	None	Full backend on personal hardware	Optional personal integrations	Lightweight web/mobile/Flutter	Recipe tracker, book log, finance tracker, home inventory, journal
AP06 Dis-tributed	Matches inner position	Matches inner position	Prototype + per-deployment instances	Same as inner position	Same as inner position	Packaged SaaS, deployable compliance tools, distributed project management

Appendix O: Naming Convention Quick Reference

Element	Convention	Rule	Examples	Anti-Examples
Entity name	Singular lowercase underscore	Singular noun or noun phrase	task, project member, cloud resource	tasks, Project-Member, cloudResource
Field name	Lowercase underscore	Descriptive noun or adjective	title, start date, is active	Title, startDate, isActive

Element	Convention	Rule	Examples	Anti-Examples
Foreign key	referenced table id	Target table name + id suffix	project id, assignee user id	proj id, assignee, fk project
Role-disambiguated FK	role table id	Role prefix + target table + id	vendor company id, client company id	company id 1, company id 2
Datetime field	* time suffix	Field name + time	created time, approved time, expires time	created at, approvedDate, timestamp
Date field	* date suffix	Field name + date	due date, birth date, effective date	dueDate, due datetime
Present-state boolean	is * prefix	is + adjective or state	is active, is published, is billable	active, published, enabled
Past-event boolean	was * prefix	was + past participle	was approved, was escalated	approved, has been approved
Governance field	prefix	Underscore + field name	requires group, access classification	requires group, accessLevel
Bridge table	source target	Both entity names singular	task label, project member, service ownership	task labels, project members
Discriminator enum	* type	entity name + type	cloud resource type, monitor type	type, kind, category
Discriminator payload	* data json	entity name + data json	cloud data json, monitor data json	data, payload, config
Versioning sibling	* version	entity name + version (auto-generated)	task version, policy version	task history, task versions
Hierarchical prefix	Specific to general	Longer names build on shorter	web site → web site widget → web site widget config	widget web site, config widget
Domain directory	NN name	Zero-padded number + domain name	01 identity, 02 core, 07 policy	identity, 2 core

Appendix P: Forbidden Pattern Quick Reference

ID	Pattern	Where	Why Forbidden	Alternative	Detection
FP01	Regex in schema	Schema YAML	Catastrophic backtracking; dialect variation	Enum sets, length bounds, anchored patterns at gate step 5	Loader rejects
FP02	Embedded logic in schema	Schema YAML	Makes validation non-deterministic	All values are literals; computed values via runners	Loader rejects
FP03	Conditional constraints in schema	Schema YAML	Cross-field logic belongs in policy evaluation	Semantic invariant policy rows at gate step 5	Loader rejects
FP04	Inheritance in schema	Schema YAML	Creates coupling between entity types	Each entity declares independently; reserved fields via opt-in	Loader rejects
FP05	Templating in schema	Schema YAML	Variation via template variables creates per-deployment divergence	One schema per AppDB; variation via runtime config data	Loader rejects
FP06	Imports within entity files	Schema YAML	Entity files must be leaf-level	Only directory.yaml imports; entity files are self-contained	Loader rejects

ID	Pattern	Where	Why Forbidden	Alternative	Detection
FP07	Field deletion	Schema evolution	Breaks version history, audit log, all consumers	Deprecate; field remains; data queryable forever	Loader rejects (diff check)
FP08	Field/entity rename	Schema evolution	Breaks every consumer by name	Add new + deprecate old; names are permanent	Loader rejects (diff check)
FP09	Type change	Schema evolution	Breaks consumers expecting prior type	Six-step duplication pattern	Loader rejects (diff check)
FP10	Range narrowing	Schema evolution	Existing rows may violate new bound	Widening only; add new field with narrower bound if needed	Loader rejects (diff check)
FP11	Enum value removal	Schema evolution	Existing rows may hold removed value	Add new field with narrower set + deprecate old	Loader rejects (diff check)
FP12	Uniqueness tightening	Schema evolution	Existing rows may violate constraint	Add new field with uniqueness + duplication pattern	Loader rejects (diff check)
FP13	Runner invoking runner	Runner design	Creates orchestrator coupling and cascading failure	Coordination through shared OpsDB state	Code review discipline

ID	Pattern	Where	Why Forbidden	Alternative	Detection
FP14	Persistent state outside OpsDB	Runner design	Other runners and queries cannot see it	Persistent state in OpsDB; in-memory for one cycle only	Code review discipline
FP15	Reinventing shared libraries	Runner design	Divergent failure modes across runners	Use standard library suite	Code review discipline
FP16	Logic in templates	Templating	Templates become opaque code	Logic in upstream runner; templates substitute values only	Code review discipline
FP17	Secrets in OpsDB	Data boundaries	OpsDB not designed for need-to-know + audit-on-read	Authority pointers to vault; library accesses at runtime	Policy enforcement
FP18	OpsDB as runtime dependency	Architecture	Services fail if OpsDB unreachable	Runners cache; local replicas; graceful degradation	Architecture review
FP19	Side tables	Schema discipline	First step toward fragmentation	Absorb into schema via new entity types	Schema steward review
FP20	Side channels	Architecture	Bypasses API; governance becomes advisory	API is only path; no exceptions	Architecture enforcement

ID	Pattern	Where	Why Forbidden	Alternative	Detection
FP21	Shared accounts	Identity	Defeats attribution	One identity per human; scoped service accounts per runner	Policy enforcement
FP22	Draft mode on compliance data	Governance flags	Creates audit gap on regulated entities	Full governance for compliance-relevant tables	Schema steward review
FP23	AI writing governed state directly	AI integration	Bypasses human oversight for consequential changes	AI proposes change sets; humans approve; AI writes observations directly	Approval rule configuration

Appendix Q: Compliance Mapping by Method

Method ID	Method Name	Compliance Properties Provided	Frameworks Satisfied
M10	Entity Design	Validated writes, typed constraints, FK integrity	SOC2-Processing-Integrity, ISO27001-A.8
M16	Governance Fields	Per-entity access control, field-level classification	SOC2-Confidentiality, PCI-DSS-7, HIPAA-Security
M19	Approval Rules	Attributed change approval, segregation of duties	SOC2-Security, SOX-IT-General-Controls

Method ID	Method Name	Compliance Properties Provided	Frameworks Satisfied
M20	Access Control	Role-based + entity-scoped + field-classified access	SOC2-Security, ISO27001-A.9, PCI-DSS-7, HIPAA-Security
M22	Retention Policies	Configurable retention per entity type per regulation	SOC2-Privacy, GDPR-Article-30
M27	Verifier Runner	Continuous evidence production with pass/fail records	PCI-DSS-11, SOC2-Availability, ISO27001-A.12
M35	Read Patterns (audit views)	Point-in-time reconstruction, change attribution	PCI-DSS-10, ISO27001-A.16
M36	Write Patterns (change sets)	Full change trail with proposer, approver, reason, timestamp	SOC2-Processing-Integrity, SOX-IT-General-Controls
M46	AI Change Proposer	AI proposals subject to same governance as human proposals	All frameworks — AI changes governed identically to human changes

Appendix R: Construction Effort Estimates by Application Type

Application Type	Position	Schema YAML (lines)	Runner Count	Runner Total (lines)	Policy Rows	Backend Effort (person-days)	Conventional Effort (person-days)	Ratio
Project management	AP01	2,000	4	800	30	15	120	8:1
CRM	AP01	3,000	5	1,200	50	20	180	9:1
HR platform	AP01	3,000	4	900	40	18	150	8:1

Application Type	Position	Schema YAML (lines)	Runner Count	Runner Total (lines)	Policy Rows	Backend Effort (person- days)	Conventional Effort (person- days)	Ratio
Compliance plat- form	AP01	2,500	4	900	60	18	135	7:1
Inventory man- age- ment	AP01	2,000	3	600	25	12	105	9:1
Personal fi- nance	AP01	1,500	3	700	10	8	60	8:1
Recipe app	AP01	1,000	1	200	5	4	45	11:1
Education plat- form	AP01	2,500	4	900	35	16	135	8:1
Legal case man- age- ment	AP01	2,500	4	900	40	16	120	8:1
Procurement	AP01	2,000	3	600	35	12	105	9:1
E-commerce	AP02	3,000	6	1,500	50	30 + hot path	240	Varies by hot path
Booking system	AP02	2,000	4	1,000	30	20 + hot path	120	Varies by hot path
CMS	AP02	2,500	4	1,000	35	22 + hot path	150	Varies by hot path
Subscription billing	AP02	3,000	6	1,500	50	28 + hot path	165	Varies by hot path
IoT fleet man- age- ment	AP02	2,500	5	1,200	40	25 + hot path	150	Varies by hot path

Application Type	Position	Schema YAML (lines)	Runner Count	Runner Total (lines)	Policy Rows	Backend Effort (person- days)	Conventional Effort (person- days)	Ratio
Chat plat- form	AP03	2,000	4	1,000	30	15 + spe- cial- ized	210	Mostly spe- cial- ized system
ML plat- form	AP03	2,500	5	1,200	35	18 + spe- cial- ized	135	Mostly spe- cial- ized system
Game back- end	AP03	2,500	5	1,200	30	18 + spe- cial- ized	180	Mostly spe- cial- ized system
Video stream- ing	AP03	2,500	5	1,200	35	18 + spe- cial- ized	195	Mostly spe- cial- ized system
Ad auc- tion	AP03	2,500	5	1,200	40	18 + spe- cial- ized	165	Mostly spe- cial- ized system

Appendix S: Method Composition by Application Type

Application Type	Domain Analysis	Schema Con- struc- tion	Policy Con- struc- tion	Runner Con- struc- tion	Frontend Integra- tion	AI As- sistance	Application Pattern
Governed- state- dominant	M01– M06, M09	M10– M18	M19– M23	M24, M25– M28, M31– M33	M34– M38	M44– M47	M39

Application Type	Domain Analysis	Schema Construction	Policy Construction	Runner Construction	Frontend Integration	AI Assistance	Application Pattern
Split-backend	M01–M09	M10–M18	M19–M23	M24–M33	M34–M38	M44–M47	M40
Operational wrapper	M01–M03, M05, M07–M09	M10–M12, M16, M18	M19–M20, M22	M24, M29–M31, M33	M34–M36, M38	M44, M46–M47	M41
Personal	M01–M04, M06	M10–M14, M17–M18	M19 (auto-approve all), M22	M24–M25, M28, M31–M32	M34–M37	M44–M45, M47	M42
Distributed	Same as inner position	Same as inner + release versioning	Same as inner + default policies	Same as inner + upgrade runner	Same as inner	Same as inner	M43 + inner pattern

Appendix T: Search API Operation Reference

Operation	Purpose	Parameters	Returns	Bounded By	Use Case
search	Query entities with filters	entity type, filters (AND/OR/NOT), fields (projection), order by, page size, cursor	Matching entity rows with projected fields	max result size, max join depth, max query time, max predicate depth, rate limit	List views, dashboards, reporting, runner get phase
get entity	Fetch single entity	entity type, entity id, join paths (optional)	Single entity with optional related entities	rate limit	Detail views, runner get phase

Operation	Purpose	Parameters	Returns	Bounded By	Use Case
get entity at time	Reconstruct state at timestamp	entity type, entity id, timestamp	Version row current at specified time	rate limit	Audit views, incident investigation, regulatory queries
get entity history	Fetch version chain	entity type, entity id, page size, cursor	Version rows in reverse chronological order	max result size, rate limit	Activity feeds, change tracking, diff views
get dependencies	Traverse relationships	entity type, entity id, join path, depth limit	Related entities along declared join path	max depth, rate limit	Dependency views, impact analysis, hierarchy traversal
search (with history)	Current state plus version chain	Same as search + view mode: with history	Entity rows with embedded version arrays	max result size, max versions per entity	Combined list + history views
search (at time)	Reconstructed state at timestamp	Same as search + view mode: at time + timestamp	Entity rows as they existed at specified time	max result size, rate limit	Historical reporting, compliance snapshots

Filter Predicates

Operator	Description	Field Types	Example
eq	Equals	All	status eq "active"
neq	Not equals	All	status neq "archived"
gt	Greater than	int, float, datetime, date	priority gt 3
gte	Greater than or equal	int, float, datetime, date	due date gte "2026-01-01"
lt	Less than	int, float, datetime, date	created time lt "2026-05-01T00:00:00Z"

Operator	Description	Field Types	Example
lte	Less than or equal	int, float, datetime, date	estimated hours lte 40.0
in	Set membership	All	status in ["open", "in progress"]
not in	Not in set	All	status not in ["archived", "deleted"]
is null	Field is null	Nullable fields	assignee user id is null
is not null	Field is not null	Nullable fields	due date is not null
starts with	Anchored prefix match	varchar	name starts with "prod-"
contains key	JSON object contains key	json	monitor data json contains key "threshold"
range	Between two values	int, float, datetime, date	priority range [1, 3]

Appendix U: Cross-Reference — Methods to Ops Book Concepts

Method	Book Concepts	Book Rules	Book Claims	Book Distinctions
M01 Entity Enumeration	C16 (Slicing the Pie), C17 (Comprehensive)	R5 (Slice the Pie)	K9 (Comprehensive → consistency)	D3 (Comprehensive vs Aggregated)
M02 Field Enumeration	C8 (Data), C7 (Knowability)	R8 (Data Primacy)	K3 (Virtual data knowable), K4 (Data > Logic)	D2 (Data vs Logic)
M04 Lifecycle Identification	C9 (Logic), C11 (System)	R8 (Data Primacy)	K7 (Tool-Logic outlives goal-Logic)	D2 (Data vs Logic)
M09 Architecture Position	C20 (Attribute Axis), C21 (Axiomatic Engineering)	R1 (90-9-0.9%)	K11 (Better not Best)	D5 (Best vs Better)
M10 Entity Design	C8 (Data), C17 (Comprehensive)	R8, R15 (Constraints On)	K4 (Data > Logic)	D2 (Data vs Logic)

Method	Book Concepts	Book Rules	Book Claims	Book Distinctions
M17 Draft Mode Config	C20 (Attribute Axis), C15 (Philosopher's Knife)	R1 (priorities), R5 (slice)	K11 (Better: explicit tradeoff)	D5 (Best vs Better)
M19 Approval Rules	C56 (AAA), C24 (Impersonal Decision)	R6 (AAA), R1 (priorities)	K1 (Control)	D12 (Insider vs Outsider)
M20 Access Control	C56 (AAA), C30 (Security Zone)	R6 (AAA), R9 (One Way)	K1 (Control)	D12 (Insider vs Outsider)
M24 Runner Design	C36 (Idempotency), C31 (Operational Logic)	R3 (Idempotency), R10 (Local Cache), R11 (Min Deps)	K14 (Idempotency → safe retry)	D4 (OpsLogic vs AppLogic)
M25 Puller Runner	C40 (Source of Truth), C41 (Knowing the Present)	R8 (Data Primacy), R12 (Verify Personally)	K15 (All data aged), K17 (Models for control sync)	D10 (Understanding vs Control Model)
M26 Reconciler Runner	C36 (Idempotency), C39 (Model for Control)	R3 (Idempotency)	K14, K17	D10 (Understanding vs Control Model)
M33 Server Runner	C35 (DOS), C34 (Distributed System)	R2 (0/1/Infinity)	K10 (No 2), K13 (Prod = one machine)	D7 (Centralized vs Decentralized)
M39 Governed-State Pattern	C17 (Comprehensive), C2 (Control)	R5, R8, R9	K1, K2, K9	D3 (Comprehensive vs Aggregated)
M44 AI Schema Generation	C8 (Data), C7 (Knowability)	R8 (Data Primacy)	K3, K4	D2 (Data vs Logic)
M46 AI Change Proposer	C56 (AAA), C2 (Control)	R6 (AAA), R3 (Idempotency)	K1 (Control), K21 (Automation amplifies)	D4 (OpsLogic vs AppLogic)

[[HOWL-COMP-9-2026](https://github.com/ghowland/papers/tree/main/papers/COMP/HOWL-COMP-9-2026)] Building Applications with OpsDB Application Architecture.
 Github: [https://github.com/ghowland/papers/tree/main/papers/COMP/HOWL-COMP-](https://github.com/ghowland/papers/tree/main/papers/COMP/HOWL-COMP-9-2026)

9-2026

[[HOWL-COMP-1-2026](#)] Implementing a Tall-Infra Data-Only Execution System .
Github: <https://github.com/ghowland/papers/tree/main/papers/COMP/HOWL-COMP-1-2026>

[[HOWL-COMP-2-2026](#)] Tall-Infra, Data-Only Development. Github: <https://github.com/ghowland/papers/tree/main/papers/COMP-2-2026>

[[HOWL-COMP-3-2026](#)] Silo OS. Github: <https://github.com/ghowland/papers/tree/main/papers/COMP/HOWL-COMP-3-2026>

[[HOWL-COMP-4-2026](#)] Geometric Security. Github: <https://github.com/ghowland/papers/tree/main/papers/COMP/HOWL-COMP-4-2026>

[[HOWL-COMP-5-2026](#)] Partial Geometric Security. Github: <https://github.com/ghowland/papers/tree/main/papers/COMP-5-2026>

[[HOWL-COMP-6-2026](#)] SIML - Silo Markup Language (TOML Format). Github: <https://github.com/ghowland/papers/tree/main/papers/COMP/HOWL-COMP-6-2026>

[[HOWL-COMP-7-2026](#)] SiQL. Github: <https://github.com/ghowland/papers/tree/main/papers/COMP/HOWL-COMP-7-2026>

[[HOWL-COMP-8-2026](#)] SiloSec Protocol Specification. Github: <https://github.com/ghowland/papers/tree/main/papers/COMP-8-2026>